

Worldvolume Hybrid Monte Carlo algorithm for group manifolds

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Abstract

The Worldvolume Hybrid Monte Carlo (WV-HMC) method [arXiv:2012.08468] is a reliable and versatile algorithm for addressing the sign problem. It resolves the ergodicity issues commonly encountered in Lefschetz thimble-based approaches at a low computational cost. In this article, as a first step toward applying WV-HMC to lattice QCD, we extend the algorithm to cases where the configuration space is a group manifold. The key idea is to formulate the algorithm using a phase-space integral over the tangent bundle of the worldvolume, which naturally carries a symplectic structure. The validity of this novel algorithm is demonstrated using the one-site model with a purely imaginary coupling constant.

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1. Introduction

The sign problem has long been a major obstacle in first-principles computations of various physical systems. While a variety of techniques have been developed to address it, they are typically tailored to specific systems of interest. Over the past fifteen years, however, there

has been a growing effort to pursue more versatile solutions. Among these, methods based on Lefschetz thimbles have attracted considerable attention owing to their mathematical rigor rooted in Picard-Lefschetz theory [1–9]. It soon became clear, however, that the original Lefschetz thimble method generally suffers from an ergodicity problem [5–7], which arises when the integration surface is deformed deeply into the complexified configuration space in order to sufficiently tame the oscillatory behavior of the path integral.

A general solution to the dilemma between reducing the sign problem and avoiding the ergodicity problem was provided by the *tempered Lefschetz thimble* (TLT) method [10–14], in which the (parallel) tempering algorithm was incorporated into the Lefschetz thimble framework by using the deformation parameter as the tempering parameter. This was the first algorithm to simultaneously resolve both the sign and ergodicity problems. Its main drawback is its high computational cost: computing the Jacobian of the deformation at each configuration exchange requires $O(N^3)$ operations, where N is the number of degrees of freedom. Moreover, the number of replicas must be increased to maintain a reasonable acceptance rate in the exchange process. This limitation was overcome by the *Worldvolume Hybrid Monte Carlo* (WV-HMC) method [15–17], in which HMC updates are performed on a continuous union of deformed surfaces (the *worldvolume*).¹ In this framework, the Jacobian need not be evaluated when generating configurations, because the path integral is formulated on the tangent bundle over the worldvolume equipped with a natural symplectic structure.

The application of Lefschetz thimble-based methods to Yang-Mills theories was first discussed in a seminal paper by Cristoforetti *et al.* [2]. However, the use of the original Lefschetz thimble method introduces the ergodicity problem. The main aim of this article is to extend the WV-HMC algorithm to group manifolds, which can be regarded as the most general setting for lattice gauge theories. We first prove Cauchy’s theorem for complexified group manifolds, then formulate the path integral over the worldvolume. This is further reformulated as an integral over the tangent bundle over the worldvolume endowed with a symplectic structure. The correctness of this novel algorithm is verified for a simple model — the one-site model. The application of the present formalism to Yang-Mills theories is straightforward, requiring no modification to the algorithmic structure, and will be presented in subsequent publications.

This paper is organized as follows. In Sect. 2, we define the complexified group $G^{\mathbb{C}}$ for a given compact group G and develop complex analysis on $G^{\mathbb{C}}$, including Cauchy’s theorem.

¹A HMC algorithm was first introduced to the Lefschetz thimble method by the Komaba group in their pioneering paper [4], where HMC updates are performed on a (single connected component of) Lefschetz thimble. A HMC algorithm on a deformed surface was proposed in Refs. [13, 14], which can be interpreted as a HMC version of the generalized thimble method [8]. We thus refer to this algorithm as *generalized thimble Hybrid Monte Carlo* (GT-HMC) in this article.

Section 3 introduces the anti-holomorphic flow, which defines the deformation of the integration surface in $G^{\mathbb{C}}$. We also provide an overview of the generalized thimble HMC (GT-HMC) and Worldvolume HMC (WV-HMC) methods for group manifolds. Before presenting a full discussion of these algorithms, we develop a general theory of Hamiltonian dynamics in $G^{\mathbb{C}}$ (Sect. 4) and on constrained submanifolds of $G^{\mathbb{C}}$ (Sect. 5). Explicit algorithms for GT-HMC and WV-HMC are then presented in Sects. 6 and 7, respectively. To validate these algorithms, we study a simple model, the one-site model, in Sect. 8. Section 9 is devoted to conclusions and future outlook. Appendix A provides a general discussion of Hamiltonian dynamics on symplectic manifolds, including an introduction to the HMC algorithm [18]. Appendix B offers a brief review of the HMC algorithm for compact groups, aiming at clarifying the correspondence between the concepts introduced in the main text and those in conventional HMC sampling on a compact group. Additional appendices contain proofs of various formulas used in the main text.

2. Complex analysis on complexified groups

In this section, we summarize aspects of complex analysis on complexified groups that will be used in subsequent sections.

2.1. Complexification of a compact group

We begin by recalling the definition of the Haar measure (dU_0) on a compact group $G = \{U_0\}$ of dimension $\dim G = N$.² Let $\text{Lie } G$ be the Lie algebra of G with a basis $\{T_a\}$ ($a = 1, \dots, N$), where the basis elements are taken to be anti-hermitian ($T_a^\dagger = -T_a$) and normalized as $\text{tr } T_a^\dagger T_b = -\text{tr } T_a T_b = \delta_{ab}$. We introduce the Maurer-Cartan 1-form on G by

$$\theta_0 \equiv dU_0 U_0^{-1} = T_a \theta_0^a \quad (\theta_0^a: \text{real 1-form}), \quad (2.1)$$

that satisfies the Maurer-Cartan equation,

$$d\theta_0 = \theta_0 \wedge \theta_0 \Leftrightarrow d\theta_0^a = \frac{1}{2} C_{bc}^a \theta_0^b \wedge \theta_0^c, \quad (2.2)$$

where C_{ab}^c are the structure constants, $[T_a, T_b] = C_{ab}^c T_c$.³ We then define the metric on G by

$$ds_0^2 \equiv \text{tr } \theta_0^\dagger \theta_0 (= -\text{tr } \theta_0 \theta_0) = (\theta_0^a)^2, \quad (2.3)$$

²We only consider a class of compact groups that can be defined uniquely in terms of unitary matrices. Their Lie algebras thus can be expressed as anti-hermitian matrices.

³Note that $C_{abc} (= -C_{ab}^c)$ is, by our convention, completely antisymmetric with respect to the index.

which represents the squared distance between U_0 , $U_0 + dU_0 \in G$. The last expression indicates that $\{\theta_0^a\}$ are the vielbeins of the metric. The Haar measure (dU_0) is then defined as the invariant volume element associated with the metric,⁴

$$(dU_0) \equiv \theta_0^1 \wedge \cdots \wedge \theta_0^N. \quad (2.4)$$

Note that the metric (and thus the Haar measure) is bi-invariant (i.e., both left- and right-invariant), $d(g_L U_0 g_R^{-1}) = (dU_0)$ ($\forall g_L, g_R \in G$), and inversion-invariant, $d(U_0^{-1}) = (dU_0)$.

The complexification $G^{\mathbb{C}}$ of G is defined as follows: We first complexify $\text{Lie } G = \bigoplus_a \mathbb{R} T_a$ to $(\text{Lie } G)^{\mathbb{C}} \equiv \bigoplus_a \mathbb{C} T_a$ as a linear space, and introduce the commutator as the natural extension of the original commutator,

$$[X + iY, X' + iY'] \equiv ([X, X'] - [Y, Y']) + i([X, Y'] + [X', Y]) \quad (X, Y, X', Y' \in \text{Lie } G). \quad (2.5)$$

The complexified group $G^{\mathbb{C}} = \{U\}$ is then defined as the set of finite products of exponentials of elements in $(\text{Lie } G)^{\mathbb{C}}$:

$$G^{\mathbb{C}} \equiv \{U = e^Z e^{Z'} \cdots e^{Z''} \mid Z, Z', \dots, Z'' \in (\text{Lie } G)^{\mathbb{C}}\}. \quad (2.6)$$

$(\text{Lie } G)^{\mathbb{C}}$ can be regarded as the (complex) Lie algebra of $G^{\mathbb{C}}$, and we denote it by $\text{Lie } G^{\mathbb{C}}$ in what follows.

We introduce the Maurer-Cartan 1-form on $G^{\mathbb{C}}$ by

$$\theta \equiv dU U^{-1} = T_a \theta^a \quad (\theta^a: \text{complex 1-form}), \quad (2.7)$$

that satisfies the Maurer-Cartan equation,

$$d\theta = \theta \wedge \theta \quad \Leftrightarrow \quad d\theta^a = \frac{1}{2} C_{bc}^a \theta^b \wedge \theta^c. \quad (2.8)$$

We then define the metric on $G^{\mathbb{C}}$ by

$$ds^2 \equiv \text{tr } \theta^\dagger \theta = \overline{\theta^a} \theta^a, \quad (2.9)$$

which represents the squared distance between U and $U + dU$ in $G^{\mathbb{C}}$. This metric is only right-invariant for $G^{\mathbb{C}}$ but still bi-invariant for G .

2.2. Derivatives of functions on $G^{\mathbb{C}}$

Any function f on $G^{\mathbb{C}}$ can be regarded as a function of U and $U^\dagger$, $f = f(U, U^\dagger)$, and we define the derivatives $Df(U, U^\dagger)$ and $\bar{D}f(U, U^\dagger) \in \text{Lie } G^{\mathbb{C}}$ from the variation:

$$\begin{aligned} \delta f(U, U^\dagger) &\equiv \text{tr} [(\delta U U^{-1}) Df(U, U^\dagger) + (\delta U U^{-1})^\dagger \bar{D}f(U, U^\dagger)] \\ &= (\delta U U^{-1})^a D_a f(U, U^\dagger) + \overline{(\delta U U^{-1})^a} \bar{D}_a f(U, U^\dagger), \end{aligned} \quad (2.10)$$

⁴We ignore the normalization of the Haar measure, which is irrelevant in the following discussions.

where⁵

$$\begin{aligned}\delta U U^{-1} &= T_a (\delta U U^{-1})^a, & Df &= T_a D^a f = -T_a D_a f, \\ (\delta U U^{-1})^\dagger &= T_a^\dagger \overline{(\delta U U^{-1})^a}, & \bar{D}f &= T_a^\dagger \bar{D}^a f = -T_a^\dagger \bar{D}_a f.\end{aligned}\quad (2.11)$$

Note that when $f(U, U^\dagger)$ is a real-valued function, we have $\bar{D}f = (Df)^\dagger$ and $\bar{D}_a f = \overline{D_a f}$. Higher derivatives are defined as $D_a D_b f \equiv D_a (D_b f)$ and so on.

We define an inner product on the space of complex matrices by

$$\langle u, v \rangle \equiv \text{Re tr } u^\dagger v (= \langle v, u \rangle = \langle u^\dagger, v^\dagger \rangle). \quad (2.12)$$

Then, for a real function $f(U, U^\dagger)$ we have the identity

$$\delta f = 2 \langle \delta U U^{-1}, (Df)^\dagger \rangle, \quad (2.13)$$

which can be shown as follows:

$$\begin{aligned}\delta f &= \text{tr} [(\delta U U^{-1}) (Df) + (\delta U U^{-1})^\dagger (Df)^\dagger] \\ &= 2 \text{Re tr } (\delta U U^{-1}) (Df) = 2 \langle \delta U U^{-1}, (Df)^\dagger \rangle.\end{aligned}\quad (2.14)$$

By replacing δ with the exterior derivative d , the above identity becomes

$$df = 2 \langle \theta, (Df)^\dagger \rangle. \quad (2.15)$$

A function $f = f(U)$ is said to be holomorphic when it is a multivariable holomorphic function of matrix elements U_{ij} . A holomorphic function $f(U)$ satisfies $\bar{D}f = 0$ and has the variation of the form

$$\delta f(U) \equiv \text{tr } (\delta U U^{-1}) Df(U) = (\delta U U^{-1})^a D_a f(U). \quad (2.16)$$

By replacing δ with the exterior derivative d , the above becomes

$$df(U) = \text{tr } \theta Df(U) = \theta^a D_a f(U). \quad (2.17)$$

A useful formula valid for $\xi \in \text{Lie } G^\mathbb{C}$ is

$$\begin{aligned}f(e^\xi U) &= f(U) + \text{tr } \xi Df(U) + O(\xi^2) \\ &= f(U) + \xi^a D_a f(U) + O(\xi^2).\end{aligned}\quad (2.18)$$

The derivatives D_a are not commutative and satisfy the following commutation relations for a holomorphic function $f(U)$:

$$D_a D_b f - D_b D_a f = -C_{ab}{}^c D_c f. \quad (2.19)$$

⁵We raise and lower indices using the rule $A^a = -A_a$ for any quantity A with index a .

This can be proved as follows. We first expand a function $f(e^\xi e^\eta U)$ with respect to ξ, η to second order:

$$\begin{aligned} f(e^\xi e^\eta U) &= f(e^\eta U) + \xi^a D_a f(e^\eta U) + \frac{1}{2} \xi^a \xi^b D_b D_a f(e^\eta U) + O(\xi^3) \\ &= f(U) + (\xi + \eta)^a D_a f(U) + [(1/2)(\xi^a \xi^b + \eta^a \eta^b) + \xi^a \eta^b] D_b D_a f(U) + \dots, \end{aligned} \quad (2.20)$$

from which we have

$$f(e^\xi e^\eta U) - f(e^\eta e^\xi U) = -\xi^a \eta^b [D_a D_b f(U) - D_b D_a f(U)] + \dots. \quad (2.21)$$

Then, by noting that the left-hand side can be rewritten to

$$\begin{aligned} f(e^{\xi+\eta+(1/2)[\xi,\eta]+\dots} U) - f(e^{\xi+\eta-(1/2)[\xi,\eta]+\dots} U) &= \text{tr} [\xi, \eta] Df(U) + \dots \\ &= \xi^a \eta^b C_{ab}^c D_c f(U) + \dots, \end{aligned} \quad (2.22)$$

and comparing the terms with coefficient $\xi^a \eta^b$ in Eqs. (2.21) and (2.22), we obtain Eq. (2.19).

For a holomorphic function $f(U)$, we define its Hessian matrix $H(U) = (H_{ab}(U))$ of size $N \times N$ by

$$H_{ab}(U) \equiv D_b D_a f(U). \quad (2.23)$$

Its action on $v = T_a v^a \in \text{Lie } G^\mathbb{C}$ is defined by

$$Hv \equiv T^a (H_{ab} v^b) = T^a v^b (D_b D_a f) = D_v Df, \quad (2.24)$$

where we have introduced $D_v \equiv v^a D_a$. The transpose $H^T(U)$ of $H(U)$ is given by

$$(H^T)_{ab}(U) = H_{ba}(U) = D_a D_b f(U), \quad (2.25)$$

and acts on v as

$$H^T v \equiv T^a (H_{ba} v^b) = T^a v^b (D_a D_b f) = DD_v f. \quad (2.26)$$

The difference between Hv and $H^T v$ can be calculated from the formula (2.19) to be⁶

$$H^T v - Hv = [Df, v]. \quad (2.27)$$

2.3. Cauchy's theorem for group manifolds

We begin by recalling Cauchy's theorem in flat space $\mathbb{C}^N = \{z = (z^i)\}$ (see Fig. 1):

Theorem 1. *Let $\mathcal{D}$ be a domain in $\mathbb{C}^N = \mathbb{R}^{2N}$ and $f(z)$ a holomorphic function on $\mathcal{D}$. Then the integral I_Σ of $f(z)$ over a real N -dimensional submanifold $\Sigma \subset \mathcal{D}$,*

$$I_\Sigma = \int_\Sigma dz f(z) \quad (dz \equiv dz^1 \wedge \dots \wedge dz^N), \quad (2.28)$$

depends only on the boundary of Σ .

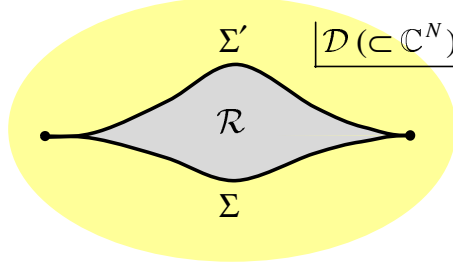


Figure 1: Cauchy's theorem for $\mathbb{C}^N$.

Proof: We set Σ and Σ' to be (oriented) real N -dimensional submanifolds of $\mathcal{D}$ with a common boundary, and $\mathcal{R}$ a region surrounded by Σ and Σ' (and thus $\partial\mathcal{R} = \Sigma' - \Sigma$). By Stokes' theorem, we have

$$I_{\Sigma'} - I_{\Sigma} = \int_{\partial\mathcal{R}} dz f(z) = \int_{\mathcal{R}} d[dz f(z)] = (-1)^N \int_{\mathcal{R}} dz \wedge df(z). \quad (2.29)$$

Since $df(z) = dz^i (\partial f(z)/\partial z^i) + d\bar{z}^i (\partial f(z)/\partial \bar{z}^i) = dz^i (\partial f(z)/\partial z^i)$, we have

$$dz \wedge df(z) = \underbrace{(dz^1 \wedge \cdots \wedge dz^N)}_{=0} \wedge dz^i \frac{\partial f(z)}{\partial z^i} = 0, \quad (2.30)$$

which means that $I_{\Sigma'} - I_{\Sigma} = 0$.

We can generalize this theorem to complexified groups as follows (see Fig. 2):

Theorem 2. Let $\mathcal{D}$ be a domain in $G^{\mathbb{C}}$ and $f(U)$ a holomorphic function on $\mathcal{D}$. Then the integral I_{Σ} of $f(U)$ over a real N -dimensional submanifold $\Sigma \subset \mathcal{D}$,

$$I_{\Sigma} = \int_{\Sigma} (dU) f(U), \quad (2.31)$$

depends only on the boundary of Σ . Here, (dU) denotes the holomorphic N -form given by

$$(dU) \equiv \theta^1 \wedge \cdots \wedge \theta^N. \quad (2.32)$$

Proof: We set Σ and Σ' to be (oriented) real N -dimensional submanifolds of $\mathcal{D}$ with a common boundary, and $\mathcal{R}$ a region surrounded by Σ and Σ' (and thus $\partial\mathcal{R} = \Sigma' - \Sigma$). By Stokes' theorem, we have

$$I_{\Sigma'} - I_{\Sigma} = \int_{\partial\mathcal{R}} (dU) f(U) = \int_{\mathcal{R}} d[(dU)f(U)]. \quad (2.33)$$

⁶This can be shown as follows: (lhs) = $T^a v^b (D_a D_b f - D_b D_a f) = -T^a v^b C_{ab}{}^c D_c f = T_a v^b (D^c f) C_{cb}{}^a = [Df, v]$.

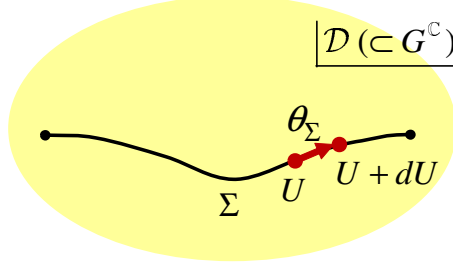


Figure 2: Cauchy's theorem for $G^{\mathbb{C}}$.

Here, from the Maurer-Cartan equation $d\theta^a = (1/2) C_{bc}^a \theta^b \wedge \theta^c$ [see Eq. (2.8)], we find that (dU) is closed, $d[(dU)] = 0$, and thus have

$$I_{\Sigma'} - I_{\Sigma} = (-1)^N \int_{\mathcal{R}} (dU) \wedge df(U). \quad (2.34)$$

Furthermore, $df(U)$ is expanded as $df(U) = \theta^a D_a f(U)$ [see Eq. (2.17)], and thus we have

$$(dU) \wedge df(U) = \underbrace{(\theta^1 \wedge \dots \wedge \theta^N)}_{=0} \wedge \theta^a D_a f(U) = 0, \quad (2.35)$$

completing the proof that $I_{\Sigma'} - I_{\Sigma} = 0$.

3. Overview of GT/WV-HMC for group manifolds

In this section, we introduce anti-holomorphic flows in the complexified group $G^{\mathbb{C}}$ and give an overview of the generalized thimble HMC (GT-HMC) and Worldvolume HMC (WV-HMC) methods, which are discussed in greater detail in subsequent sections. Our main goal is to show that the path integrals in both GT-HMC and WV-HMC can be reformulated as integrals over the tangent bundles of the configuration spaces, which naturally carry symplectic structures.

3.1. Anti-holomorphic flows in $G^{\mathbb{C}}$

We consider the expectation value of an observable $\mathcal{O}(U_0)$ defined through a path integral over a compact group $G = \{U_0\}$ as

$$\langle \mathcal{O} \rangle \equiv \frac{\int_G (dU_0) e^{-S(U_0)} \mathcal{O}(U_0)}{\int_G (dU_0) e^{-S(U_0)}}. \quad (3.1)$$

We complexify G to $G^{\mathbb{C}} = \{U\}$ and assume that both $e^{-S(U)}$ and $e^{-S(U)} \mathcal{O}(U)$ are holomorphic on $G^{\mathbb{C}}$ (which holds in most cases of interest). By Cauchy's theorem, the expression

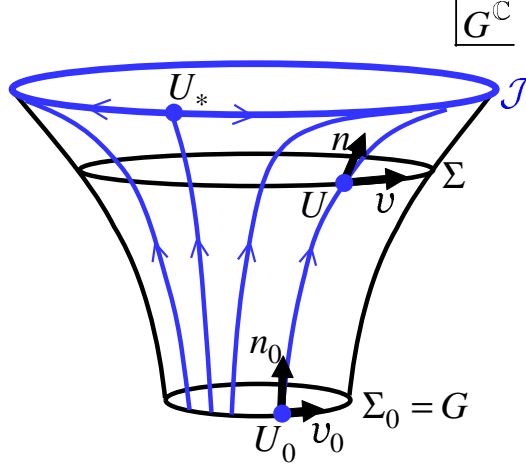


Figure 3: Deformation of $\Sigma_0 = G$ into $G^{\mathbb{C}}$. The deformed surface Σ approaches Lefschetz thimble $\mathcal{J}$, which consists of points flowing out from a critical point U_* . v_0 (or n_0) is a tangent (or normal) vector at $U_0 \in \Sigma_0$, while v (or n) is a tangent (or normal) vector at $U \in \Sigma$.

above can be rewritten as a ratio of integrals over a new integration surface Σ that is obtained by a continuous deformation of G (see Fig. 3):

$$\langle \mathcal{O} \rangle = \frac{\int_{\Sigma} (dU) e^{-S(U)} \mathcal{O}(U)}{\int_{\Sigma} (dU) e^{-S(U)}}. \quad (3.2)$$

Thus, even when the original path integral on $\Sigma_0 = G$ suffers from the severe sign problem due to the highly oscillatory behavior of $e^{-i \text{Im} S(U_0)}$, this problem can be significantly alleviated if $\text{Im} S(U)$ is almost constant on the new integration surface Σ .

Such deformation is given by the anti-holomorphic flow equation

$$\dot{U} = \xi(U) U \quad (3.3)$$

with

$$\xi(U) \equiv [DS(U)]^{\dagger}. \quad (3.4)$$

This leads to the (in)equality

$$[S(U)]^{\cdot} = \text{tr}[(\dot{U} U^{-1}) DS(U)] = \text{tr}[(DS(U))^{\dagger} (DS(U))] \geq 0, \quad (3.5)$$

which shows that the real part $\text{Re} S(U)$ always increase along the flow (except at critical points where $DS(U) = 0$), while the imaginary part $\text{Im} S(U)$ remains constant along the flow. The Lefschetz thimble $\mathcal{J}$ associated with a critical point U_* is defined as the set of points flowing out of U_* (see Fig. 3). $\text{Im} S(U)$ is constant over $\mathcal{J}$ because of the invariance of $\text{Im} S(U)$ along the flow, $\text{Im} S(U) = \text{Im} S(U_*)$ ($U \in \mathcal{J}$). Thus, the oscillatory behavior

of the integrand is expected to be much relaxed if we deform the integration surface with sufficiently large flow time t so that the deformed surface comes to the vicinity of $\mathcal{J}$.

As we will see in later sections, in order to sample Σ , we need to transport a tangent vector $v_0 \in T_{U_0}\Sigma_0$ ($\Sigma_0 = G$) to a tangent vector $v \in T_U\Sigma$ (see Fig. 3).⁷ The transformation rule can be found by considering an adjacent flow of the form $e^v U$,

$$(e^v U)^\bullet = [DS(e^v U)]^\dagger e^v U, \quad (3.6)$$

and extracting terms linear in v , which leads to

$$\dot{v} = (Hv)^\dagger + [\xi(U), v] \quad \text{with} \quad v|_{t=0} = v_0. \quad (3.7)$$

Here, H is the Hessian matrix of $S(U)$, $Hv = D_v DS = T^a H_{ab} v^b$ with $H_{ab} = D_b D_a S$ [see Eq. (2.24)]. We can also introduce the flow of a normal vector $n = n_t \in N_U \Sigma_t$ such that it stays orthogonal to $v = v_t \in T_U \Sigma_t$ at any flow time t :

$$\dot{n} = -(H^T n)^\dagger - [\xi^\dagger, n] \quad \text{with} \quad n|_{t=0} = n_0, \quad (3.8)$$

where H^T is the transpose of H , $H^T n = DD_n S = T^a H_{ba} n^b$ [see Eq. (2.26)]. We actually can show that if $\langle n_0, v_0 \rangle = 0$, then $\langle n, v \rangle = 0$ holds for all t . In fact,

$$\langle n, v \rangle^\bullet = (\text{Re tr } n^\dagger v)^\bullet = 0, \quad (3.9)$$

because

$$\begin{aligned} (\text{tr } n^\dagger v)^\bullet &= \text{tr } (\dot{n}^\dagger v + n^\dagger \dot{v}) \\ &= \text{tr } [(-H^T n + [\xi, n^\dagger]) v + n^\dagger ((Hv)^\dagger + [\xi, v])] \\ &= \text{tr } [-(H^T n) v + n^\dagger (Hv)^\dagger + [\xi, n^\dagger v]] \\ &= -2i \text{Im tr } n (Hv) \in i\mathbb{R}. \end{aligned} \quad (3.10)$$

The flow equations (3.7) and (3.8) define $\mathbb{R}$ -linear maps $E : T_{U_0}\Sigma_0 \rightarrow T_U\Sigma$ and $F : N_{U_0}\Sigma_0 \rightarrow N_U\Sigma$ (see Fig. 4), which together define the $\mathbb{R}$ -linear map $A : T_{U_0}G^\mathbb{C} \rightarrow T_U G^\mathbb{C}$ by

$$A : T_{U_0}G^\mathbb{C} \ni w_0 = v_0 + n_0 \mapsto w = v + n = Ev_0 + Fn_0 \in T_U G^\mathbb{C}. \quad (3.11)$$

⁷ $T_U \Sigma$ denotes the tangent space at U , and $N_U \Sigma$ the normal space at U , $N_U \Sigma = \{n \in T_U G^\mathbb{C} \mid \langle n, v \rangle = 0 \ (\forall v \in T_U \Sigma)\}$.

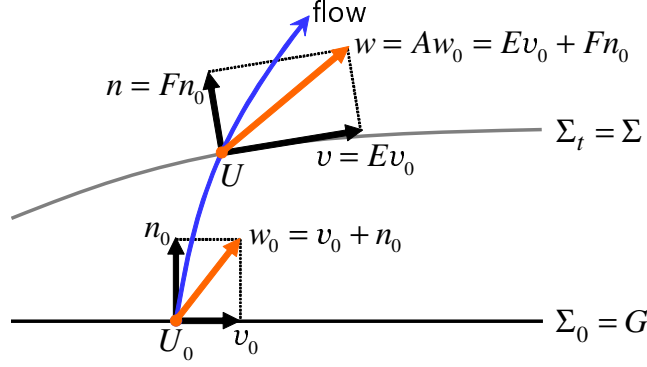


Figure 4: Linear transformation: $w_0 = v_0 + n_0 \mapsto w = Ev_0 + Fn_0$.

3.2. Overview of GT-HMC for group manifolds

We start from the following expression [Eq. (3.2)]:

$$\langle \mathcal{O} \rangle = \frac{\int_{\Sigma} (dU)_{\Sigma} e^{-S(U)} \mathcal{O}(U)}{\int_{\Sigma} (dU)_{\Sigma} e^{-S(U)}}, \quad (3.12)$$

where $\Sigma \equiv \Sigma_t$ denotes the deformed surface at flow time t . To clarify notation, we denote the Maurer-Cartan form on Σ and the corresponding holomorphic N -form as $\theta_{\Sigma} \equiv \theta|_{\Sigma} = T_a \theta_{\Sigma}^a$ and $(dU)_{\Sigma} \equiv (dU)|_{\Sigma} = \theta_{\Sigma}^1 \wedge \cdots \wedge \theta_{\Sigma}^N$, respectively. The one-forms θ_{Σ} are linear in θ_0 and take the form

$$\theta_{\Sigma}^a = E_b^a \theta_0^b, \quad (3.13)$$

from which we obtain the relation

$$(dU)_{\Sigma} = \det E \cdot (dU_0). \quad (3.14)$$

The Jacobian matrix $E = (E_b^a)$ can be computed by integrating the flow equations for the vectors $E_a \equiv T_b E_a^b$ [see Eq. (3.7)]:

$$\dot{E}_a = (HE_a)^{\dagger} + [\xi, E_a] \quad \text{with} \quad E_a|_{t=0} = T_a. \quad (3.15)$$

Since θ_{Σ} can be expressed as

$$\theta_{\Sigma} = E_a \theta_0^a, \quad (3.16)$$

the induced metric on Σ is given by

$$ds_{\Sigma}^2 = \langle \theta_{\Sigma}, \theta_{\Sigma} \rangle = \gamma_{ab} \theta_0^a \theta_0^b \quad (3.17)$$

with

$$\gamma_{ab} \equiv \langle E_a, E_b \rangle = \text{Re tr } E_a^{\dagger} E_b, \quad (3.18)$$

which represents the squared distance between U , $U + dU \in \Sigma$. This metric defines the invariant volume element on Σ ,

$$|dU|_\Sigma = \sqrt{\gamma} (dU_0). \quad (3.19)$$

One can check that $\text{Im tr } E_a^\dagger E_b$ does not vanish in general (as opposed to the flat case [4]). This implies that one cannot write γ_{ab} as $\text{tr } E_a^\dagger E_b = \overline{E_a^c} E_b^c$, and thus $\sqrt{\gamma}$ does not agree with $|\det E|$ generically (i.e., $(dU)_\Sigma / |dU|_\Sigma = \det E / \sqrt{\gamma}$ is not a pure phase factor).

We write the expectation value (3.12) as a ratio of reweighted averages on Σ :

$$\langle \mathcal{O} \rangle = \frac{\langle \mathcal{F}(U) \mathcal{O}(U) \rangle_\Sigma}{\langle \mathcal{F}(U) \rangle_\Sigma}. \quad (3.20)$$

Here, $\langle \cdots \rangle_\Sigma$ is defined by

$$\langle g(U) \rangle_\Sigma \equiv \frac{\int_\Sigma |dU|_\Sigma e^{-\text{Re } S(U)} g(U)}{\int_\Sigma |dU|_\Sigma e^{-\text{Re } S(U)}}, \quad (3.21)$$

and $\mathcal{F}(U)$ is the associated reweighting factor:

$$\mathcal{F}(U) \equiv \frac{(dU)_\Sigma}{|dU|_\Sigma} e^{-i \text{Im } S(U)} = \frac{\det E}{\sqrt{\gamma}} e^{-i \text{Im } S(U)}. \quad (3.22)$$

The reweighted average $\langle \cdots \rangle_\Sigma$ can be expressed as a path integral over the phase space by rewriting the measure $|dU|_\Sigma = \sqrt{\gamma} (dU_0)$ to the form [17]

$$|dU|_\Sigma \propto (dU_0) (dp) e^{-(1/2) \gamma^{ab} p_a p_b}, \quad (3.23)$$

where $(\gamma^{ab}) \equiv (\gamma_{ab})^{-1}$. Furthermore, the phase-space measure $(dU_0) (dp) \equiv (\prod_a \theta_0^a) (\prod_a dp_a)$ can be rewritten as

$$d\Omega_\Sigma \equiv \frac{\omega_\Sigma^N}{N!} \quad (3.24)$$

with the symplectic 2-form

$$\omega_\Sigma \equiv d(p_a \theta_0^a) = dp_a \wedge \theta_0^a + \frac{1}{2} C_{bc}^a \pi_a \theta_0^b \wedge \theta_0^c. \quad (3.25)$$

Note that terms with $\theta_0^b \wedge \theta_0^c$ drop out in ω_Σ^N . We thus obtain the phase-space path integral in the parameter-space representation:

$$\langle g(U) \rangle_\Sigma = \frac{\int d\Omega_\Sigma e^{-(1/2) \gamma^{ab} p_a p_b - \text{Re } S(U(U_0))} g(U(U_0))}{\int d\Omega_\Sigma e^{-(1/2) \gamma^{ab} p_a p_b - \text{Re } S(U(U_0))}}. \quad (3.26)$$

For Monte Carlo calculations, it is more convenient to rewrite everything in terms of the target space coordinates U . To do this, we introduce the momentum $\pi = T_a \pi^a \in T_U \Sigma$ as

$$\pi \equiv E_a p^a \quad (p^a \equiv \gamma^{ab} p_b) \quad \Leftrightarrow \quad \pi^a = E_b^a p^b, \quad (3.27)$$

which satisfies the identity⁸

$$\langle \pi, \pi \rangle = \gamma^{ab} p_a p_b \quad (\pi \in T_U \Sigma). \quad (3.28)$$

One can also show that the 1-form

$$a_\Sigma \equiv \langle \pi, \theta_\Sigma \rangle = \text{Re tr } \pi^\dagger \theta_\Sigma \quad (3.29)$$

can be expressed as $a_\Sigma = p_a \theta_0^a$, and thus we find that a_Σ is a symplectic potential of ω_Σ , $\omega_\Sigma = da_\Sigma$, which implies

$$\omega_\Sigma = da_\Sigma = d\langle \pi, \theta_\Sigma \rangle. \quad (3.30)$$

We thus have succeeded in rewriting Eq. (3.26) to a ratio of path integrals over the tangent bundle over Σ , $T\Sigma \equiv \{(U, \pi) \mid U \in \Sigma, \pi \in T_U \Sigma\}$:

$$\langle g(U) \rangle_\Sigma = \frac{\int_{T\Sigma} d\Omega_\Sigma e^{-H(U, \pi)} g(U)}{\int_{T\Sigma} d\Omega_\Sigma e^{-H(U, \pi)}}, \quad (3.31)$$

where the Hamiltonian $H(U, \pi)$ takes the form⁹

$$H(U, \pi) = \frac{1}{2} \langle \pi, \pi \rangle + V(U, U^\dagger) \quad (3.32)$$

with the (real-valued) potential

$$V(U, U^\dagger) = \text{Re } S(U) = \frac{1}{2} [S(U) + \overline{S(U)}]. \quad (3.33)$$

An explicit algorithm for generating configurations $(U, \pi) \in T\Sigma$ from the distribution $\propto e^{-H(U, \pi)}$ is given in Sect. 6 after we establish a general theory of Hamiltonian dynamics on constrained submanifolds of $G^\mathbb{C}$.

3.3. Overview of WV-HMC for group manifolds

As mentioned in Sect. 1, the GT-HMC algorithm often introduces the ergodicity problem when the flow time is taken sufficiently large to reduce the oscillatory behavior of $e^{-i \text{Im } S}$. This trade-off between reducing the sign problem and maintaining ergodicity can be resolved at a low computational cost by employing the WV-HMC method [15], in which the path integral over a single deformed surface Σ [Eq. (3.2)] is replaced by an integral over a continuous union of deformed surfaces (the worldvolume).

⁸This can be shown as $\langle \pi, \pi \rangle = \langle E_a, E_b \rangle p^a p^b = \gamma_{ab} p^a p^b = \gamma^{ab} p_a p_b$.

⁹A more precise expression is $H(U, U^\dagger, \pi, \pi^\dagger) = (1/2) \langle \pi, \pi \rangle + V(U, U^\dagger)$, but we abbreviate it as in the main text to simplify expressions.

Although the method was originally introduced for the flat case, it can be generalized to group manifolds as follows. We first note that when we set the deformed surface to $\Sigma = \Sigma_t$ (deformed surface at flow time t), both the numerator and the denominator of Eq. (3.2) are independent of t due to Cauchy's theorem, so that we can take an average over t separately with an arbitrary common weight $e^{-W(t)}$ as in Ref. [15]:

$$\langle \mathcal{O} \rangle = \frac{\int_{\Sigma_t} (dU)_{\Sigma_t} e^{-S(U)} \mathcal{O}(U)}{\int_{\Sigma_t} (dU)_{\Sigma_t} e^{-S(U)}} = \frac{\int dt e^{-W(t)} \int_{\Sigma_t} (dU)_{\Sigma_t} e^{-S(U)} \mathcal{O}(U)}{\int dt e^{-W(t)} \int_{\Sigma_t} (dU)_{\Sigma_t} e^{-S(U)}}. \quad (3.34)$$

This can be regarded as a ratio of integrals over the *worldvolume* $\mathcal{R}$ defined by

$$\mathcal{R} \equiv \bigcup_t \Sigma_t = \{U(t, U_0) \in G^{\mathbb{C}} \mid U_0 \in G, t \in \mathbb{R}\}, \quad (3.35)$$

where $U(t, U_0)$ denotes the flowed configuration at flow time t from the initial configuration U_0 .

One can restrict the extent of $\mathcal{R}$ in the t -direction to a finite interval $\sim [T_0, T_1]$ by appropriately choosing the function $W(t)$ [17, 19], whose possible form is given in Sect. 7.3. The lower bound T_0 is chosen such that there are no ergodicity issues on Σ_{T_0} . The upper bound T_1 is chosen such that oscillatory integrals are sufficiently tamed there, which can be monitored by calculating average reweighting factor $\langle \mathcal{F}(U) \rangle_{\Sigma_t}$ for various t with GT-HMC as in Ref. [15]. After global equilibrium is well established across $\mathcal{R}$, we estimate the expectation value $\langle \mathcal{O} \rangle$ using sample averages over a subinterval $[\tilde{T}_0, \tilde{T}_1]$ ($T_0 \leq \tilde{T}_0 < \tilde{T}_1 \leq T_1$), which avoids both the sign problem near T_0 and possible geometric complexities near T_1 .¹⁰

The invariant volume element $|dU|_{\mathcal{R}}$ of $\mathcal{R}$ can be expressed as follows. We first introduce the Maurer-Cartan form on $\mathcal{R}$ using the parametrization $U = U(t, U_0)$:

$$\theta_{\mathcal{R}} = dU U^{-1}|_{\mathcal{R}} = \xi dt + \theta_{\Sigma}, \quad (3.36)$$

where we have used the flow equation, $\dot{U} = \xi U$. By decomposing the vector ξ into tangential and normal components as $\xi = \xi_v + \xi_n$ with $\xi_v \in T_U \Sigma$ and $\xi_n \in N_U \Sigma$, we obtain the orthogonal decomposition of $\theta_{\mathcal{R}}$:

$$\theta_{\mathcal{R}} = \xi_n dt + (\theta_{\Sigma} + \xi_v dt). \quad (3.37)$$

This leads to the induced metric on $\mathcal{R}$ expressed in the ADM decomposition:

$$ds_{\mathcal{R}}^2 = \langle \theta_{\mathcal{R}}, \theta_{\mathcal{R}} \rangle = \alpha^2 dt^2 + \gamma_{ab} (\theta_{\Sigma}^a + \beta^a dt) (\theta_{\Sigma}^b + \beta^b dt), \quad (3.38)$$

where the induced metric γ_{ab} , the shift vector β^a , and the lapse function α are defined by

$$\gamma_{ab} = \langle E_a, E_b \rangle, \quad \beta^a = \gamma^{ab} \langle E_b, \xi_v \rangle, \quad \alpha^2 = \langle \xi_n, \xi_n \rangle. \quad (3.39)$$

¹⁰The subinterval $[\tilde{T}_0, \tilde{T}_1]$ is determined by the condition that the estimate of $\mathcal{O}$ varies only within small statistical errors against small changes of the subinterval [15].

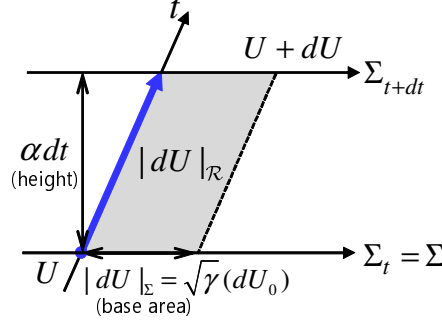


Figure 5: Invariant volume element $|dU|_{\mathcal{R}}$ of worldvolume $\mathcal{R}$.

The corresponding invariant measure on $\mathcal{R}$ is then given by (see Fig. 5)

$$|dU|_{\mathcal{R}} = \alpha dt |dU|_{\Sigma} = \alpha \sqrt{\gamma} dt (dU_0). \quad (3.40)$$

Meanwhile, the holomorphic N -form is given by $(dU)_{\Sigma} = \det E \cdot (dU_0)$ [Eq. (3.14)]. Substituting this into Eq. (3.34), the path integral can be rewritten as

$$\langle \mathcal{O} \rangle = \frac{\int_{\mathcal{R}} |dU|_{\mathcal{R}} e^{-V(U)} \mathcal{F}(U) \mathcal{O}(U)}{\int_{\mathcal{R}} |dU|_{\mathcal{R}} e^{-V(U)} \mathcal{F}(U)} = \frac{\langle \mathcal{F}(U) \mathcal{O}(U) \rangle_{\mathcal{R}}}{\langle \mathcal{F}(U) \rangle_{\mathcal{R}}}. \quad (3.41)$$

Here, $V(U)$ and $\mathcal{F}(U)$ denote the potential and the associated reweighting factor, respectively.¹¹

$$V(U) \equiv \text{Re } S(U) + W(t(U, U^{\dagger})), \quad (3.42)$$

$$\mathcal{F}(U) \equiv \frac{dt (dU)_{\Sigma}}{|dU|_{\mathcal{R}}} e^{-i \text{Im } S(U)} = \alpha^{-1} \frac{\det E}{\sqrt{\gamma}} e^{-i \text{Im } S(U)}, \quad (3.43)$$

and $\langle \cdots \rangle_{\mathcal{R}}$ represents the reweighted average on $\mathcal{R}$ defined by

$$\langle g(U) \rangle_{\mathcal{R}} \equiv \frac{\int_{\mathcal{R}} |dU|_{\mathcal{R}} e^{-V(U)} g(U)}{\int_{\mathcal{R}} |dU|_{\mathcal{R}} e^{-V(U)}}. \quad (3.44)$$

As in GT-HMC, the reweighted averages $\langle \cdots \rangle_{\mathcal{R}}$ can be recast as path integrals over the phase space. To this end, we introduce a basis of $T_U \mathcal{R}$, $\{\hat{E}_{\mu} = T_a \hat{E}_{\mu}^a\}$ ($\mu = 0, 1, \dots, N$), with $\hat{E}_0 \equiv \xi$ and $\hat{E}_a \equiv E_a = T_b E_a^b$ ($a = 1, \dots, N$). Then, $\theta_{\mathcal{R}}$ is expanded as

$$\theta_{\mathcal{R}} = \xi dt + E_a \theta_0^a = \hat{E}_{\mu} \hat{\theta}_0^{\mu} \quad (3.45)$$

with $\hat{\theta}_0^0 \equiv dt$ and $\hat{\theta}_0^a \equiv \theta_0^a$ ($a = 1, \dots, N$), from which the induced metric $ds_{\mathcal{R}}^2$ is expressed as

$$ds_{\mathcal{R}}^2 = \hat{\gamma}_{\mu\nu} \hat{\theta}_0^{\mu} \hat{\theta}_0^{\nu} \quad (3.46)$$

¹¹The real function $t(U, U^{\dagger})$ returns the flow time t for configuration $U = U(t, U_0)$.

with

$$\hat{\gamma}_{\mu\nu} \equiv \langle \hat{E}_\mu, \hat{E}_\nu \rangle. \quad (3.47)$$

Thus, the invariant volume element $|dU|_{\mathcal{R}}$ is rewritten to

$$|dU|_{\mathcal{R}} = \sqrt{\hat{\gamma}} \prod_{\mu=0}^N \hat{\theta}_0^\mu \left(= \alpha \sqrt{\gamma} dt (dU_0) \right), \quad (3.48)$$

which can be further rewritten to the following form by introducing the momentum $\hat{p}_\mu$ ($\mu = 0, 1, \dots, N$):

$$|dU|_{\mathcal{R}} = \left(\prod_{\mu=0}^N \hat{\theta}_0^\mu \right) \left(\prod_{\mu=0}^N d\hat{p}_\mu \right) e^{-(1/2) \hat{\gamma}^{\mu\nu} \hat{p}_\mu \hat{p}_\nu}, \quad (3.49)$$

where $(\hat{\gamma}^{\mu\nu}) \equiv (\hat{\gamma}_{\mu\nu})^{-1}$. Furthermore, the phase-space volume $(\prod_{\mu=0}^N \hat{\theta}_0^\mu) (\prod_{\mu=0}^N d\hat{p}_\mu) = dt (\prod_{a=1}^N \theta_0^a) (\prod_{\mu=0}^N d\hat{p}_\mu)$ can be expressed as

$$d\Omega_{\mathcal{R}} \equiv \frac{\omega_{\mathcal{R}}^N}{(N+1)!} \quad (3.50)$$

with the symplectic 2-form $\omega_{\mathcal{R}}$ defined by

$$\begin{aligned} \omega_{\mathcal{R}} &\equiv d(\hat{p}_\mu \hat{\theta}_0^\mu) = d\hat{p}_0 \wedge dt + d(\hat{p}_a \theta_0^a) \\ &= d\hat{p}_0 \wedge dt + d\hat{p}_a \wedge \theta_0^a + \frac{1}{2} C_{bc}^a \hat{p}_a \theta_0^b \wedge \theta_0^c. \end{aligned} \quad (3.51)$$

Note that terms with $\theta_0^b \wedge \theta_0^c$ drop out in $\omega_{\mathcal{R}}^{N+1}$. We thus obtain the phase-space path integral in the parameter-space representation:

$$\langle g(U) \rangle_{\mathcal{R}} = \frac{\int d\Omega_{\mathcal{R}} e^{-(1/2) \hat{\gamma}^{\mu\nu} \hat{p}_\mu \hat{p}_\nu - V(U(t, U_0))} g(U(t, U_0))}{\int d\Omega_{\mathcal{R}} e^{-(1/2) \hat{\gamma}^{\mu\nu} \hat{p}_\mu \hat{p}_\nu - V(U(t, U_0))}}. \quad (3.52)$$

As in the flat case [15, 17], for practical Monte Carlo simulations, it is more convenient to rewrite everything in terms of the target space coordinates U . To this end, we introduce the momentum $\pi = T_a \pi^a \in T_U \mathcal{R}$ by

$$\pi \equiv \hat{E}_\mu \hat{p}^\mu \quad (\hat{p}^\mu \equiv \hat{\gamma}^{\mu\nu} \hat{p}_\nu) \quad \Leftrightarrow \quad \pi^a = \hat{E}_\mu^a \hat{p}^\mu, \quad (3.53)$$

which satisfies the identity

$$\langle \pi, \pi \rangle = \hat{\gamma}^{\mu\nu} \hat{p}_\mu \hat{p}_\nu \quad (\pi \in T_U \mathcal{R}). \quad (3.54)$$

One can also show that the 1-form

$$a_{\mathcal{R}} \equiv \langle \pi, \theta_{\mathcal{R}} \rangle = \text{Re tr } \pi^\dagger \theta_{\mathcal{R}} \quad (3.55)$$

can be expressed as $a_{\mathcal{R}} = \hat{p}_\mu \hat{\theta}_0^\mu$. This means that $a_{\mathcal{R}}$ is a symplectic potential of $\omega_{\mathcal{R}}$, $\omega_{\mathcal{R}} = da_{\mathcal{R}}$, and we obtain the identity

$$\omega_{\mathcal{R}} = d \langle \pi, \theta_{\mathcal{R}} \rangle. \quad (3.56)$$

We thus have succeeded in rewriting Eq. (3.52) to a ratio of path integrals over the tangent bundle over $\mathcal{R}$, $T\mathcal{R} = \{(U, \pi) \mid U \in \mathcal{R}, \pi \in T_U \mathcal{R}\}$:

$$\langle g(U) \rangle_{\mathcal{R}} = \frac{\int_{T\mathcal{R}} d\Omega_{\mathcal{R}} e^{-H(U, \pi)} g(U)}{\int_{T\mathcal{R}} d\Omega_{\mathcal{R}} e^{-H(U, \pi)}}, \quad (3.57)$$

where the Hamiltonian $H(U, \pi)$ takes the form

$$H(U, \pi) = \frac{1}{2} \langle \pi, \pi \rangle + V(U, U^\dagger) \quad (3.58)$$

with the (real-valued) potential

$$V(U, U^\dagger) = \text{Re } S(U) + W(t(U, U^\dagger)). \quad (3.59)$$

An explicit algorithm to generate configurations $(U, \pi) \in T\mathcal{R}$ from the distribution $\propto e^{-H(U, \pi)}$ is presented in Sect. 7 after we establish a general theory of Hamiltonian dynamics on constrained submanifolds of $G^{\mathbb{C}}$.

4. Unconstrained Hamiltonian dynamics in $G^{\mathbb{C}}$

Although in the context of GT/WV-HMC our main interest lies in Hamiltonian dynamics on a constrained submanifold of $G^{\mathbb{C}}$, we begin by considering the unconstrained Hamiltonian dynamics to clarify the mathematical structure that remains unchanged by the introduction of constraints. For a review of the HMC algorithm for a compact group G , presented in a manner consistent with this section, see Appendix B.

4.1. Continuum version

Our phase space is given by the tangent bundle over $G^{\mathbb{C}}$, $TG^{\mathbb{C}} = \{(U, \pi) \mid U \in G^{\mathbb{C}}, \pi \in T_U G^{\mathbb{C}}\}$. We define the symplectic potential a by

$$a \equiv \langle \pi, \theta \rangle = \text{Re tr } \pi^\dagger \theta = \frac{1}{2} \text{tr } (\pi^\dagger \theta + \pi \theta^\dagger), \quad (4.1)$$

with which the symplectic 2-form ω and the volume $4N$ -form $d\Omega$ are obtained as (recall that $\dim_{\mathbb{R}} G^{\mathbb{C}} = 2N$):

$$\omega = da, \quad d\Omega = \frac{1}{(2N)!} \omega^{2N}. \quad (4.2)$$

We restrict our attention to Hamiltonians of the form

$$H(U, \pi) = K(\pi, \pi^\dagger) + V(U, U^\dagger) \quad \text{with} \quad K(\pi, \pi^\dagger) = \frac{1}{2} \langle \pi, \pi \rangle. \quad (4.3)$$

To *define* consistent dynamics with a given Hamiltonian $H(U, \pi)$, we consider a trajectory $(U(s), \pi(s))$ parametrized by a continuous parameter s , and define it as a stationary path of the action of the first-order form:

$$I[U(s), \pi(s)] = \int ds [\langle \pi, \dot{U} U^{-1} \rangle - H(U, \pi)], \quad (4.4)$$

where $\dot{U} \equiv dU/ds$. The equations of motion follow from the variation of the action,¹²

$$\begin{aligned} \delta I[U, \pi] = \frac{1}{2} \int ds \operatorname{tr} \big\{ & \delta \pi^\dagger [\dot{U} U^{-1} - \pi] + \delta \pi [(\dot{U} U^{-1})^\dagger - \pi^\dagger] \\ & + (\delta U U^{-1}) [-\dot{\pi}^\dagger + [\pi, \pi^\dagger] - 2 DV] \\ & + (\delta U U^{-1})^\dagger [-\dot{\pi} + [\pi, \pi^\dagger] - 2 (DV)^\dagger] \big\}, \end{aligned} \quad (4.5)$$

to be

$$\dot{U} = \pi U, \quad (4.6)$$

$$\dot{\pi} = -2 [DV(U, U^\dagger)]^\dagger + [\pi, \pi^\dagger]. \quad (4.7)$$

Note the appearance of the term $[\pi, \pi^\dagger]$, which is not present in Hamiltonian dynamics for G (where $\pi^\dagger = -\pi$).

Hamilton's equations (4.6) and (4.7) satisfy the following three key properties:

(1) Reversibility: When a trajectory $(U(s), \pi(s))$ ($0 \leq s \leq 1$) is a solution to Hamilton's equations, its time-reversal,

$$U(s) \rightarrow \tilde{U}(s) \equiv U(1-s), \quad (4.8)$$

$$\pi(s) \rightarrow \tilde{\pi}(s) \equiv -\pi(1-s), \quad (4.9)$$

is also a solution. In fact, we have

$$\dot{\tilde{U}}(s) = -\dot{U}(1-s) = -\pi(1-s) U(1-s) = \tilde{\pi}(s) \tilde{U}(s), \quad (4.10)$$

and

$$\begin{aligned} \dot{\tilde{\pi}}(s) &= \dot{\pi}(1-s) \\ &= -2 [DV(U(1-s), U^\dagger(1-s))]^\dagger + [\pi(1-s), \pi^\dagger(1-s)] \\ &= -2 [DV(\tilde{U}(s), \tilde{U}^\dagger(s))]^\dagger + [\tilde{\pi}(s), \tilde{\pi}^\dagger(s)]. \end{aligned} \quad (4.11)$$

¹²We have neglected boundary terms and used the identity

$$\operatorname{tr} \pi^\dagger \delta(\dot{U} U^{-1}) = \operatorname{tr} \delta U U^{-1} (-\dot{\pi}^\dagger + [\pi, \pi^\dagger]) + (\text{total derivatives}).$$

(2) Symplecticity:

$$\overset{\circ}{\omega} = 0. \quad (4.12)$$

In fact, explicit calculations show¹³

$$\overset{\circ}{a} = \langle \pi, \theta \rangle^\circ = \langle \overset{\circ}{\pi}, \theta \rangle + \langle \pi, \overset{\circ}{\theta} \rangle = d\left(\frac{1}{2} \text{tr } \pi^\dagger \pi - V\right), \quad (4.13)$$

and thus we have

$$\overset{\circ}{\omega} = d\overset{\circ}{a} = 0. \quad (4.14)$$

(3) Energy conservation:

$$\overset{\circ}{H} = 0. \quad (4.15)$$

The first-order action (4.4) determines the symplectic potential to be $a = \langle \pi, \theta \rangle$. When G is $SU(n)$ or $U(n)$, the Poisson brackets take the following form:¹⁴

$$\{U_{ij}, U_{kl}\} = 0, \quad (4.16)$$

$$\{U_{ij}, U_{kl}^\dagger\} = 0, \quad (4.17)$$

$$\{U_{ij}, \pi_{kl}^\dagger\} = 2 \left(\delta_{il} U_{kj} - \frac{1}{n} U_{ij} \delta_{kl} \right), \quad (4.18)$$

$$\{\pi_{ij}^\dagger, U_{kl}\} = 2 \left(-U_{il} \delta_{kj} + \frac{1}{n} \delta_{ij} U_{kl} \right), \quad (4.19)$$

$$\{U_{ij}^\dagger, \pi_{kl}\} = 2 \left(U_{il}^\dagger \delta_{kj} - \frac{1}{n} U_{ij}^\dagger \delta_{kl} \right), \quad (4.20)$$

$$\{\pi_{ij}, U_{kl}^\dagger\} = 2 \left(-\delta_{il} U_{kj}^\dagger + \frac{1}{n} \delta_{ij} U_{kl}^\dagger \right), \quad (4.21)$$

$$\{\pi_{ij}, \pi_{kl}\} = 2 \left(-\delta_{il} \pi_{kj} + \pi_{il} \delta_{kj} \right), \quad (4.22)$$

$$\{\pi_{ij}^\dagger, \pi_{kl}^\dagger\} = 2 \left(\delta_{il} \pi_{kj}^\dagger - \pi_{il}^\dagger \delta_{kj} \right), \quad (4.23)$$

$$\{\pi_{ij}, \pi_{kl}^\dagger\} = 0. \quad (4.24)$$

¹³Note that $\overset{\circ}{\theta} = (dUU^{-1})^\circ = d\pi - [\theta, \pi]$.

¹⁴This can be obtained in parallel to the derivation given in Appendix B.2. The $-(1/n)$ terms are not present when $G = U(n)$.

Using the Poisson brackets, one can easily show the following identities:

$$\{U, K\} = \pi U, \quad (4.25)$$

$$\{U, V\} = 0, \quad (4.26)$$

$$\{\pi, K\} = [\pi, \pi^\dagger], \quad (4.27)$$

$$\{\pi, V\} = -2(DV)^\dagger, \quad (4.28)$$

$$\{U^\dagger, K\} = U^\dagger \pi^\dagger, \quad (4.29)$$

$$\{U^\dagger, V\} = 0, \quad (4.30)$$

$$\{\pi^\dagger, K\} = [\pi, \pi^\dagger], \quad (4.31)$$

$$\{\pi^\dagger, V\} = -2DV, \quad (4.32)$$

which reproduce Hamilton's equations:

$$\{U, H\} = \pi U, \quad \{\pi, H\} = -2(DV)^\dagger + [\pi, \pi^\dagger], \quad (4.33)$$

$$\{U^\dagger, H\} = U^\dagger \pi^\dagger, \quad \{\pi^\dagger, H\} = -2DV + [\pi, \pi^\dagger]. \quad (4.34)$$

We now make an important observation that will play a crucial role in the discussions to follow. From Eqs. (4.27) and (4.31), we have

$$\{\pi, K\} = [\pi, \pi^\dagger] = [\pi - \pi^\dagger, \pi], \quad (4.35)$$

$$\{\pi^\dagger, K\} = [\pi, \pi^\dagger] = [\pi - \pi^\dagger, \pi^\dagger]. \quad (4.36)$$

This implies that the Poisson bracket $\{*, K\}$ acts as the commutator $[\pi - \pi^\dagger, *]$ when applied to functions depending only on π and $\pi^\dagger$:

$$\{f(\pi, \pi^\dagger), K\} = [\pi - \pi^\dagger, f(\pi, \pi^\dagger)]. \quad (4.37)$$

4.2. Discrete version

We discretize the “transfer matrix” $T_H = e^{\epsilon\{*, H\}}$ with step size $\Delta s = \epsilon$ in the symmetric form:

$$T \equiv T_{V/2} T_K T_{V/2}, \quad (4.38)$$

whose deviation from T_H is $O(\epsilon^3)$, $T = T_H + O(\epsilon^3)$.¹⁵

¹⁵For notational simplicity, we denote $e^{\epsilon\{*, f\}}$ by T_f (rather than $T_{\epsilon f}$). The relation $T = T_H + O(\epsilon^3)$ follows directly from the identity $e^{(\epsilon/2)A} e^{\epsilon B} e^{(\epsilon/2)B} = e^{\epsilon(A+B)} + O(\epsilon^3)$, which holds for linear operators A and B of any type.

Straightforward computation yields the following identities (see Appendix D for a proof):

$$T_{V/2}U = U, \quad (4.39)$$

$$T_{V/2}\pi = \pi - \epsilon [DV(U, U^\dagger)]^\dagger \quad (\equiv \pi_{1/2}), \quad (4.40)$$

$$T_K U = e^{\epsilon(\pi - \pi^\dagger)} e^{\epsilon\pi^\dagger} U, \quad (4.41)$$

$$T_K \pi = e^{\epsilon(\pi - \pi^\dagger)} \pi e^{-\epsilon(\pi - \pi^\dagger)}. \quad (4.42)$$

We thus find that the transformed variables $U' \equiv TU$ and $\pi' \equiv T\pi$ are given by

$$\begin{aligned} U' &= T_{V/2} T_K T_{V/2} U = T_{V/2} T_K U \\ &= T_{V/2} \{e^{\epsilon(\pi - \pi^\dagger)} e^{\epsilon\pi^\dagger} U\} \\ &= e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} e^{\epsilon\pi_{1/2}^\dagger} U, \end{aligned} \quad (4.43)$$

and

$$\begin{aligned} \pi' &= T_{V/2} T_K T_{V/2} \pi = T_{V/2} T_K (\pi - \epsilon [DV(U, U^\dagger)]^\dagger) \\ &= T_{V/2} \{e^{\epsilon(\pi - \pi^\dagger)} \pi e^{-\epsilon(\pi - \pi^\dagger)} - \epsilon [DV(e^{\epsilon(\pi - \pi^\dagger)} e^{\epsilon\pi^\dagger} U, (e^{\epsilon(\pi - \pi^\dagger)} e^{\epsilon\pi^\dagger} U)^\dagger)]^\dagger\} \\ &= e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} \pi_{1/2} e^{-\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} - \epsilon [DV(U', U'^\dagger)]^\dagger. \end{aligned} \quad (4.44)$$

This gives an explicit three-step update algorithm for Hamiltonian dynamics on $TG^\mathbb{C}$:

$$\pi_{1/2} = \pi - \epsilon [DV(U, U^\dagger)]^\dagger, \quad (4.45)$$

$$U' = e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} e^{\epsilon\pi_{1/2}^\dagger} U, \quad (4.46)$$

$$\pi' = e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} \pi_{1/2} e^{-\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} - \epsilon [DV(U', U'^\dagger)]. \quad (4.47)$$

The update (4.45)–(4.47) satisfies the following three properties, just as the standard leapfrog algorithm does in flat space:

(1) Exact reversibility: The transformation is invariant under time reversal,

$$U \rightarrow \tilde{U} \equiv U', \quad (4.48)$$

$$\pi \rightarrow \tilde{\pi} \equiv -\pi', \quad (4.49)$$

$$\pi_{1/2} \rightarrow \tilde{\pi}_{1/2} \equiv -e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} \pi_{1/2} e^{-\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)}, \quad (4.50)$$

$$U' \rightarrow \tilde{U}' \equiv U, \quad (4.51)$$

$$\pi \rightarrow \tilde{\pi}' \equiv -\pi. \quad (4.52)$$

(2) Symplecticity: The symplectic form ω (and thus the volume form $d\Omega = \omega^{2N}/(2N)!$) is preserved,

$$\omega' = \omega. \quad (4.53)$$

(3) Approximate energy conservation:

$$H(U', \pi') = H(U, \pi) + O(\epsilon^3). \quad (4.54)$$

Among these, only the first property is nontrivial. In fact, property (2) follows directly from the fact that the transformation T is composed of canonical transformations of the form $T_f = e^{\epsilon \{*, f\}}$. This can also be verified explicitly by noting that

$$T_{V/2} a = a - \frac{\epsilon}{2} dV(U, U^\dagger), \quad (4.55)$$

$$T_K a = a + \frac{\epsilon}{2} d(\text{tr } \pi^\dagger \pi), \quad (4.56)$$

which implies $T_{V/2} \omega = \omega$, $T_K \omega = \omega$ and thus $T\omega = \omega$. Property (3) holds because $T = T_H + O(\epsilon^3)$ and $TH = T_H H + O(\epsilon^3) = H + O(\epsilon^3)$. To demonstrate property (1), we first note the relation

$$\tilde{\pi}_{1/2} - \tilde{\pi}_{1/2}^\dagger = -(\pi_{1/2} - \pi_{1/2}^\dagger), \quad (4.57)$$

which can be proved as

$$(\text{lhs}) = -e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} (\pi_{1/2} - \pi_{1/2}^\dagger) e^{-\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} = (\text{rhs}). \quad (4.58)$$

This relation allows Eq. (4.50) to be rewritten as

$$\pi_{1/2} = -e^{\epsilon(\tilde{\pi}_{1/2} - \tilde{\pi}_{1/2}^\dagger)} \tilde{\pi}_{1/2} e^{-\epsilon(\tilde{\pi}_{1/2} - \tilde{\pi}_{1/2}^\dagger)}. \quad (4.59)$$

Then, by using Eq. (4.45) we have

$$\begin{aligned} \tilde{\pi}' &= -\pi \\ &= -\pi_{1/2} - \epsilon [DV(U, U^\dagger)]^\dagger \\ &= e^{\epsilon(\tilde{\pi}_{1/2} - \tilde{\pi}_{1/2}^\dagger)} \tilde{\pi}_{1/2} e^{-\epsilon(\tilde{\pi}_{1/2} - \tilde{\pi}_{1/2}^\dagger)} - \epsilon [DV(\tilde{U}', \tilde{U}'^\dagger)]^\dagger, \end{aligned} \quad (4.60)$$

which is the time-reversed form of Eq. (4.47). Similarly, using Eq. (4.47), we obtain

$$\begin{aligned} \tilde{\pi}_{1/2} &= -e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} \pi_{1/2} e^{-\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} \\ &= -\pi' - \epsilon [DV(U', U'^\dagger)]^\dagger \\ &= \tilde{\pi} - \epsilon [DV(\tilde{U}, \tilde{U}^\dagger)]^\dagger, \end{aligned} \quad (4.61)$$

which is the time-reversed form of Eq. (4.45). To show the remaining part, we note the

relation¹⁶

$$e^{-\epsilon \pi_{1/2}^\dagger} e^{-\epsilon (\pi_{1/2} - \pi_{1/2}^\dagger)} = e^{\epsilon (\tilde{\pi}_{1/2} - \tilde{\pi}_{1/2}^\dagger)} e^{\epsilon \tilde{\pi}_{1/2}^\dagger}. \quad (4.62)$$

Then, we have

$$\begin{aligned} \tilde{U}' &= U \\ &= e^{-\epsilon \pi_{1/2}^\dagger} e^{-\epsilon (\pi_{1/2} - \pi_{1/2}^\dagger)} U' \\ &= e^{\epsilon (\tilde{\pi}_{1/2} - \tilde{\pi}_{1/2}^\dagger)} e^{\epsilon \tilde{\pi}_{1/2}^\dagger} \tilde{U}, \end{aligned} \quad (4.63)$$

which is the time-reversed form of Eq. (4.46).

5. Constrained Hamiltonian dynamics in $G^\mathbb{C}$

In this section, we develop a general framework for Hamiltonian dynamics on a real m -dimensional submanifold $\mathcal{S}$ of $G^\mathbb{C}$. Specifically, we set $\mathcal{S} = \Sigma$ with $m = N$ for GT-HMC (Sect. 6), and $\mathcal{S} = \mathcal{R}$ with $m = N + 1$ for WV-HMC (Sect. 7).

5.1. Continuum version

We assume that the submanifold $\mathcal{S}$ is defined by $2N - m$ independent real-valued functions $\phi^r(U) = 0$ ($r = 1, \dots, 2N - m$). The phase space is given by the tangent bundle over $\mathcal{S}$, $T\mathcal{S} \equiv \{(U, \pi) \mid U \in \mathcal{S}, \pi \in T_U \mathcal{S}\}$, with symplectic potential $a_\mathcal{S}$ and symplectic form $\omega_\mathcal{S} = da_\mathcal{S}$ defined by

$$a_\mathcal{S} = \langle \pi, \theta_\mathcal{S} \rangle, \quad (5.1)$$

$$\omega_\mathcal{S} = d\langle \pi, \theta_\mathcal{S} \rangle, \quad (5.2)$$

where $\theta_\mathcal{S} = (dUU^{-1})|_\mathcal{S}$ is the Maurer-Cartan form on $\mathcal{S}$. The phase-space volume form is given by

$$d\Omega_\mathcal{S} \equiv \frac{\omega_\mathcal{S}^m}{m!}. \quad (5.3)$$

¹⁶This can be proved as

$$\begin{aligned} e^{-\epsilon (\tilde{\pi}_{1/2} - \tilde{\pi}_{1/2}^\dagger)} e^{-\epsilon \pi_{1/2}^\dagger} e^{-\epsilon (\pi_{1/2} - \pi_{1/2}^\dagger)} &= e^{\epsilon (\pi_{1/2} - \pi_{1/2}^\dagger)} e^{-\epsilon \pi_{1/2}^\dagger} e^{-\epsilon (\pi_{1/2} - \pi_{1/2}^\dagger)} \\ &= \exp[-\epsilon e^{\epsilon (\pi_{1/2} - \pi_{1/2}^\dagger)} \pi_{1/2}^\dagger e^{-\epsilon (\pi_{1/2} - \pi_{1/2}^\dagger)}] \\ &= e^{\epsilon \tilde{\pi}_{1/2}^\dagger}. \end{aligned}$$

To define consistent Hamiltonian dynamics on $T\mathcal{S}$ with a Hamiltonian of the form

$$H(U, \pi) = \frac{1}{2} \langle \pi, \pi \rangle + V(U, U^\dagger), \quad (5.4)$$

we consider the first-order action

$$I[U, \pi, \lambda] = \int ds \left[\langle \pi, \dot{U} U^{-1} \rangle - H(U, \pi) - \lambda_r \phi^r(U) \right], \quad (5.5)$$

where $\dot{U} \equiv dU/ds$ and $\lambda_r \in \mathbb{R}$ are Lagrange multipliers. The resulting equations of motion are

$$\dot{U} = \pi U, \quad (5.6)$$

$$\dot{\pi} = -2 [DV(U, U^\dagger)]^\dagger - 2 \lambda_r [D\phi^r(U, U^\dagger)]^\dagger \quad (5.7)$$

subject to the constraints¹⁷

$$\phi^r(U) = 0, \quad (5.8)$$

$$\langle \pi, (D\phi^r)^\dagger \rangle = 0. \quad (5.9)$$

It is easy to verify that the symplectic potential a changes as $\dot{a} = d[(1/2)\langle \pi, \pi \rangle - V(U) - \lambda_r \phi^r(U)]$, from which it follows that $\dot{\omega} = d\dot{a} = 0$. Furthermore, we define

$$\lambda_{\text{cont}} \equiv \lambda_r (D\phi^r)^\dagger, \quad (5.10)$$

which lies in the normal space $N_U \mathcal{S}$.¹⁸ It then follows that $\dot{H} = -2 \langle \pi, \lambda_{\text{cont}} \rangle = 0$.

5.2. Discrete version

A discretized version of Eqs. (5.6)–(5.9) with step size $\Delta s = \epsilon$ is obtained by modifying Eqs. (4.45)–(4.47) using the RATTLE algorithm [20, 21] as follows:

$$\pi_{1/2} = \pi - \epsilon [DV(U, U^\dagger)]^\dagger - \epsilon \lambda, \quad (5.11)$$

$$U' = e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} e^{\epsilon \pi_{1/2}^\dagger} U, \quad (5.12)$$

$$\pi' = e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} \pi_{1/2} e^{-\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} - \epsilon [DV(U', U'^\dagger)]^\dagger - \epsilon \lambda'. \quad (5.13)$$

¹⁷The second constraint follows from $0 = \dot{\phi}^r(U) = 2 \langle \dot{U} U^{-1}, (D\phi^r)^\dagger \rangle = 2 \langle \pi, (D\phi^r)^\dagger \rangle$ [see Eq. (2.14)].

¹⁸In fact, for any vector $v \in T_U \mathcal{S}$, we have

$$\langle \lambda_{\text{cont}}, v \rangle = \lambda_r \text{Re} \langle (D\phi^r)^\dagger, v \rangle = (\lambda_r/2) \text{tr} [v D\phi^r + v^\dagger (D\phi^r)^\dagger] = (\lambda_r/2) \lim_{\epsilon \rightarrow 0} (1/\epsilon) [\phi^r(e^{\epsilon v} U) - \phi^r(U)] = 0.$$

Here, the Lagrange multipliers $\lambda \in N_U \mathcal{S}$ and $\lambda' \in N_{U'} \mathcal{S}$ are determined, respectively, such that $U' \in \mathcal{S}$ and $\pi' \in T_{U'} \mathcal{S}$. If one introduces the projector $\Pi_{\mathcal{S}}$ (*res.* $\Pi'_{\mathcal{S}}$) from $T_U G^{\mathbb{C}}$ (*res.* $T_{U'} G^{\mathbb{C}}$) onto $T_U \mathcal{S}$ (*res.* $T_{U'} \mathcal{S}$), then the determination of λ' is replaced by the projection:

$$\pi' = \Pi'_{\mathcal{S}} \tilde{\pi}', \quad (5.14)$$

where

$$\tilde{\pi}' \equiv e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} \pi_{1/2} e^{-\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} - \epsilon [DV(U', U'^\dagger)]^\dagger. \quad (5.15)$$

The update (5.11)–(5.13) satisfies the following three properties, just as in the unconstrained case:

(1) Exact reversibility: The transformation is invariant under time reversal,

$$U \rightarrow \tilde{U} \equiv U', \quad (5.16)$$

$$\pi \rightarrow \tilde{\pi} \equiv -\pi', \quad (5.17)$$

$$\pi_{1/2} \rightarrow \tilde{\pi}_{1/2} \equiv -e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} \pi_{1/2} e^{-\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)}, \quad (5.18)$$

$$U' \rightarrow \tilde{U}' \equiv U, \quad (5.19)$$

$$\pi \rightarrow \tilde{\pi}' \equiv -\pi, \quad (5.20)$$

$$\lambda \rightarrow \tilde{\lambda} \equiv \lambda', \quad (5.21)$$

$$\lambda' \rightarrow \tilde{\lambda}' \equiv \lambda. \quad (5.22)$$

(2) Symplecticity: The symplectic form $\omega_{\mathcal{S}}$ (and thus the volume form $d\Omega_{\mathcal{S}} = \omega_{\mathcal{S}}^m/m!$) is preserved,

$$\omega'_{\mathcal{S}} = \omega_{\mathcal{S}}. \quad (5.23)$$

(3) Approximate energy conservation:

$$H(U', \pi') = H(U, \pi) + O(\epsilon^3). \quad (5.24)$$

Property (1) can be proved in a way similar to the unconstrained case. Property (2) can also be proved in a similar way. The only difference is that an extra term $-\epsilon \langle \lambda, (dUU^{-1})_{\mathcal{R}} \rangle$ is added to the right-hand side of Eq. (4.55), but this term vanishes because $\lambda \in N_U \mathcal{R}$. To prove property (3), we introduce intermediate variables,

$$\bar{\pi} \equiv \pi_{1/2} + \epsilon [DV(U, U^\dagger)]^\dagger (= \pi - \epsilon \lambda), \quad (5.25)$$

$$\bar{\pi}' \equiv e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} \pi e^{-\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} - \epsilon [DV(U', U'^\dagger)]^\dagger (= \pi' + \epsilon \lambda'). \quad (5.26)$$

We then have

$$\pi_{1/2} = \bar{\pi} - \epsilon [DV(U, U^\dagger)]^\dagger \quad (5.27)$$

$$U' = e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} e^{\epsilon\pi_{1/2}^\dagger} U, \quad (5.28)$$

$$\bar{\pi}' = e^{\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} \pi e^{-\epsilon(\pi_{1/2} - \pi_{1/2}^\dagger)} - \epsilon [DV(U', U'^\dagger)]^\dagger. \quad (5.29)$$

This has the same form as Eqs. (4.45)–(4.47), and thus we have

$$H(U', \bar{\pi}') = H(U, \bar{\pi}) + O(\epsilon^3). \quad (5.30)$$

Furthermore, we have

$$\begin{aligned} H(U, \bar{\pi}) &= H(U, \pi - \epsilon\lambda) = \frac{1}{2} \langle \pi - \epsilon\lambda, \pi - \epsilon\lambda \rangle + V(U, U^\dagger) \\ &= H(U, \pi) + \frac{\epsilon^2}{2} \langle \lambda, \lambda \rangle \\ &= H(U, \pi) + \frac{\epsilon^2}{2} \langle \lambda_{\text{cont}}, \lambda_{\text{cont}} \rangle + O(\epsilon^3), \end{aligned} \quad (5.31)$$

and

$$\begin{aligned} H(U', \bar{\pi}') &= H(U', \pi' + \epsilon\lambda') = \frac{1}{2} \langle \pi' + \epsilon\lambda', \pi' + \epsilon\lambda' \rangle + V(U', U'^\dagger) \\ &= H(U', \pi') + \frac{\epsilon^2}{2} \langle \lambda', \lambda' \rangle \\ &= H(U', \pi') + \frac{\epsilon^2}{2} \langle \lambda_{\text{cont}}, \lambda_{\text{cont}} \rangle + O(\epsilon^3). \end{aligned} \quad (5.32)$$

Here, we have used $\langle \pi, \lambda \rangle = \langle \pi', \lambda' \rangle = 0$, and the expansions $\lambda = \lambda_{\text{cont}} + O(\epsilon)$, $\lambda' = \lambda_{\text{cont}} + O(\epsilon)$, where λ_{cont} is the Lagrange multiplier in the continuum version [Eq. (5.10)]. We thus have

$$H(U', \pi') - H(U, \pi) = H(U', \bar{\pi}') - H(U, \bar{\pi}) + O(\epsilon^3) = O(\epsilon^3). \quad (5.33)$$

5.3. HMC for a submanifold $\mathcal{S}$ of $G^\mathbb{C}$

Our aim is to numerically evaluate the reweighted average $\langle \cdots \rangle_{\mathcal{S}}$ that is expressed as a path integral over the tangent bundle $T\mathcal{S}$,

$$\langle g(U) \rangle_{\mathcal{S}} = \frac{\int_{T\mathcal{S}} d\Omega_{\mathcal{S}} e^{-H(U, \pi)} g(U)}{\int_{T\mathcal{S}} d\Omega_{\mathcal{S}} e^{-H(U, \pi)}}. \quad (5.34)$$

This can be estimated by a sample average over configurations $\{(U, \pi)\}$ drawn from the distribution $\propto e^{-H(U, \pi)}$. In the remainder of this article, we rescale the Lagrange multipliers

as $\lambda \rightarrow \lambda/\epsilon^2$ and $\lambda' \rightarrow \lambda'/\epsilon^2$ to simplify later expressions, which means that λ and λ' are $O(\epsilon^2)$ quantities. We also introduce an intermediate variable

$$Z \equiv \epsilon\pi - \epsilon^2 [DV(U, U^\dagger)], \quad (5.35)$$

so that

$$\pi_{1/2} = (Z - \lambda)/\epsilon. \quad (5.36)$$

To construct a Markov chain with equilibrium distribution proportional to $e^{-H(U, \pi)}$ with H satisfying $H(U, -\pi) = H(U, \pi)$, we define a set of stochastic processes $P_{(k)}(U', \pi' | U, \pi)$ ($k = 1, \dots, K$) such that each satisfies the detailed balance condition of molecular dynamics,

$$P_{(k)}(U', \pi' | U, \pi) e^{-H(U, \pi)} = P_{(k)}(U, -\pi | U', -\pi') e^{-H(U', -\pi')}. \quad (5.37)$$

We also assume that the product $P \equiv P_{(K)} \cdots P_{(1)}$ has a proper ergodicity property that ensures unique convergence. Under these conditions, it can be shown that $e^{-H(U, \pi)}$ is the unique eigenvector of P with unit eigenvalue,

$$\int_{TS} d\Omega'_S P(U, \pi | U', \pi') e^{-H(U', \pi')} = e^{-H(U, \pi)}, \quad (5.38)$$

and thus $e^{-H(U, \pi)} / \int_{TS} d\Omega_S e^{-H(U, \pi)}$ is the equilibrium distribution reached by the Markov chain defined by P .

In the present case, we consider the following two stochastic processes:

(1) Heat bath for π :

$$P_{(1)}(U', \pi' | U, \pi) = e^{-(1/2) \langle \pi', \pi' \rangle} \delta_S(U', U), \quad (5.39)$$

where $\delta_S(U', U)$ is the delta function on $\mathcal{S}$.¹⁹ π' can be generated by drawing $\tilde{\pi}'$ in $G^{\mathbb{C}}$ from the Gaussian distribution $e^{-(1/2) \langle \tilde{\pi}', \tilde{\pi}' \rangle}$ (see Appendix C) and projecting it onto $T_U \mathcal{S}$ as

$$\pi' = \Pi'_S \tilde{\pi}'. \quad (5.40)$$

This process obviously satisfies the detailed balance condition (5.37).

(2) Molecular dynamics (MD) plus Metropolis test:

$$P_{(2)}(U', \pi' | U, \pi) = \min(1, e^{[H(U', \pi') - H(U, \pi)]}) \delta_{TS}((U', \pi'), T^{N_{\text{MD}}}(U, \pi)) \quad \text{when } (U', \pi') \neq (U, \pi). \quad (5.41)$$

¹⁹When $U \in \mathcal{S}$ is parametrized as $U = U(t, U_0)$ as in GT/WV-HMC, the delta function is proportional to $\delta(U'_0, U_0)$ for GT-HMC ($\mathcal{S} = \Sigma$), and to $\delta(t' - t) \delta(U'_0, U_0)$ for WV-HMC ($\mathcal{S} = \mathcal{R}$), where $\delta(U'_0, U_0)$ is the bi-invariant delta function associated with the Haar measure of G . Jacobian factors can be ignored in the argument for detailed balance because $U' = U$ in Eq. (5.37).

One MD step $T : (U, \pi) \mapsto (U', \pi')$ is defined as follows:

$$Z = \epsilon\pi - \epsilon^2 [DV(U, U^\dagger)]^\dagger, \quad (5.42)$$

$$U' = e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U, \quad (5.43)$$

$$\tilde{\pi}' = e^{(Z-\lambda)-(Z-\lambda)^\dagger} \frac{1}{\epsilon} (Z - \lambda) e^{-(Z-\lambda)+(Z-\lambda)^\dagger} - \epsilon [DV(U', U'^\dagger)]^\dagger, \quad (5.44)$$

$$\pi' = \Pi'_S \tilde{\pi}', \quad (5.45)$$

where $\lambda \in N_U \mathcal{S}$ is determined so that $U' \in \mathcal{S}$. N_{MD} is the number of MD steps. $\delta_{TS}(U, \pi)$ is the delta function with respect to the volume form $d\Omega_S$. The volume preservation implies that the Jacobian of the map $(U, \pi) \mapsto (U', \pi')$ is unity. The reversibility together with the unit Jacobian then gives the identity

$$\delta_{TS}((U', \pi'), T^{N_{\text{MD}}}(U, \pi)) = \delta_{TS}((U, -\pi), T^{N_{\text{MD}}}(U', -\pi')), \quad (5.46)$$

which ensures that $P_{(2)}$ satisfies the detailed balance condition (5.37).

Since the observables of our interest depend only on U , the algorithm can be viewed as a stochastic process on U alone:

- Step 1 (momentum refresh): With a given $U \in \mathcal{S}$, generate $\tilde{\pi} \in \text{Lie } G^\mathbb{C}$ from a Gaussian, and project it to get $\pi = \Pi_S \tilde{\pi} \in T_U \mathcal{S}$.
- Step 2 (molecular dynamics): Perform N_{MD} MD steps using the procedure above [Eqs. (5.42)–(5.45)].
- Step 3 (Metropolis test): Accept U' with probability $\min(1, e^{-[H(U', \pi') - H(U, \pi)]})$.

The only remaining ingredient in the whole algorithm are the construction of the projector Π_S and the determination of the Lagrange multiplier $\lambda \in N_U \mathcal{S}$, which depend on details of the geometry of $\mathcal{S}$. We explain the case $\mathcal{S} = \Sigma$ (GT-HMC) in Sect. 6 and explain the case $\mathcal{S} = \mathcal{R}$ (WV-HMC) in Sect. 7.

6. Algorithm of GT-HMC for group manifolds

In the GT-HMC algorithm, we consider the submanifold $\mathcal{S} = \Sigma$, which is a deformed surface at flow time t (i.e., $\Sigma = \Sigma_t$).

6.1. Projector in GT-HMC

We first describe an algorithm to project a vector $w \in T_U G^\mathbb{C}$ onto the tangent space $T_U \Sigma$ and the normal space $N_U \Sigma$:

$$w = v + n \quad (v = \Pi_\Sigma w \in T_U \Sigma, \quad n = (1 - \Pi_\Sigma)w \in N_U \Sigma). \quad (6.1)$$

Algorithm 1 Orthogonal decomposition of $w \in T_U G^{\mathbb{C}}$ into $v \in T_U \Sigma$ and $n \in N_U \Sigma$

- 1: Solve the linear system $Aw_0 = w$ for w_0 .
 - 2: Decompose w_0 into $v_0 = (1/2)(w_0 - w_0^\dagger)$ and $n_0 = (1/2)(w_0 + w_0^\dagger)$.
 - 3: Compute $v = Ev_0$ and $n = Fn_0$ by integrating Eqs. (3.7) and (3.8).
-

By using the $\mathbb{R}$ -linear map $A : T_{U_0} G^{\mathbb{C}} \ni w_0 = v_0 + n_0 \mapsto w = v + n = Ev_0 + Fn_0 \in T_U G^{\mathbb{C}}$ [Eq. (3.11)], the decomposition (6.1) for a given $w \in T_U G^{\mathbb{C}}$ can be carried out by Algorithm 1. Note that, if an iterative method (such as BiCGStab) is used to solve $Aw_0 = w$ in Step 1, Steps 2 and 3 are no longer necessary as in the flat case [9]. This is because, during the iteration, one repeatedly computes $E\tilde{v}_0$ and $F\tilde{n}_0$ for a trial solution $\tilde{w}_0 = \tilde{v}_0 + \tilde{n}_0$, so that $v = Ev_0$ and $n = Fn_0$ are already obtained upon convergence.

6.2. Constrained molecular dynamics on Σ

We begin by recalling the quantity we aim to evaluate [Eq. (3.20)]:

$$\langle \mathcal{O} \rangle = \frac{\langle \mathcal{F}(U) \mathcal{O}(U) \rangle_\Sigma}{\langle \mathcal{F}(U) \rangle_\Sigma}. \quad (6.2)$$

Here, the reweighted average on Σ is given by [Eq. (3.31)]:

$$\langle g(U) \rangle_\Sigma = \frac{\int_{T\Sigma} d\Omega_\Sigma e^{-H(U,\pi)} g(U)}{\int_{T\Sigma} d\Omega_\Sigma e^{-H(U,\pi)}}, \quad (6.3)$$

where [Eqs. (3.24), (3.30), (3.32), (3.33)]

$$d\Omega_\Sigma = \frac{\omega_\Sigma^N}{N!} \quad \text{with} \quad \omega_\Sigma = d\langle \pi, \theta_\Sigma \rangle, \quad (6.4)$$

$$H(U, \pi) = \frac{1}{2} \langle \pi, \pi \rangle + V(U, U^\dagger) \quad \text{with} \quad V(U, U^\dagger) = \text{Re } S(U). \quad (6.5)$$

The reweighting factor $\mathcal{F}(U)$ is given by [Eq. (3.22)]:

$$\mathcal{F}(U) = \frac{\det E}{\sqrt{\gamma}} e^{-i \text{Im } S(U)}. \quad (6.6)$$

Since $[DV(U, U^\dagger)]^\dagger = (1/2) [DS(U)]^\dagger = (1/2) \xi(U)$, a single MD step is given by

$$Z = \epsilon \pi - (\epsilon^2/2) \xi(U), \quad (6.7)$$

$$U' = e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U, \quad (6.8)$$

$$\tilde{\pi}' = e^{(Z-\lambda)-(Z-\lambda)^\dagger} \frac{1}{\epsilon} (Z - \lambda) e^{-(Z-\lambda)+(Z-\lambda)^\dagger} - \frac{\epsilon}{2} \xi(U'), \quad (6.9)$$

$$\pi' = \Pi'_\Sigma \tilde{\pi}'. \quad (6.10)$$

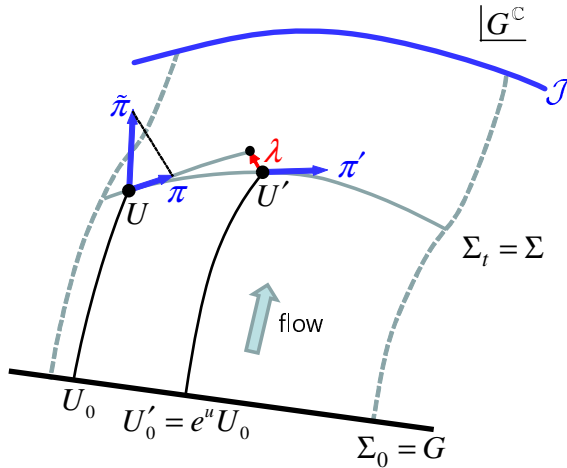


Figure 6: RATTLE in GT-HMC: $(U, \pi) \rightarrow (U', \pi')$. The initial momentum π in the molecular dynamics is constructed by projecting $\tilde{\pi} \in T_U G^{\mathbb{C}}$ onto $T_U \Sigma$, where $\tilde{\pi}$ is randomly generated with the Gaussian distribution $\propto e^{-(1/2)\langle \tilde{\pi}, \tilde{\pi} \rangle}$.

Here, the Lagrange multiplier $\lambda \in N_U \Sigma$ is determined such that $U' \in \Sigma$, which is equivalent to finding a pair $(u \in T_{U_0} \Sigma_0, \lambda \in N_U \Sigma)$ satisfying the following equation for a given $U = U(t, U_0)$ (see Fig. 6):

$$e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U(t, U_0) = U(t, e^u U_0). \quad (6.11)$$

We solve this equation by using Newton's method based on the ansatz that an update $u \leftarrow u + \Delta u$ and $\lambda \leftarrow \lambda + \Delta \lambda$ can be effectively performed by solving the equation

$$e^{-\Delta\lambda} e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U(t, U_0) = U(t, e^{\Delta u} e^u U_0). \quad (6.12)$$

We linearize this by writing $e^{-\Delta\lambda} \approx 1 - \Delta\lambda$ and $U(t, e^{\Delta u} e^u U_0) \approx (1 + E_{\text{new}} \Delta u) U(t, e^u U_0)$, where E_{new} is the $\mathbb{R}$ -linear map $E_{\text{new}} : T_{e^u U_0} \Sigma_0 \rightarrow T_{U_{\text{new}}} \Sigma$ with $U_{\text{new}} \equiv U(t, e^u U_0)$. The algorithm then proceeds as follows:

Step 1: Initialize $\Delta u = 0$ and $\Delta \lambda = 0$.

Step 2: Solve the linearized equation

$$E_{\text{new}}\Delta u + \Delta\lambda e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} UU_{\text{new}}^{-1} = e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} UU_{\text{new}}^{-1} - 1, \quad (6.13)$$

where $U = U(t, U_0)$ and $U_{\text{new}} = U(t, e^u U_0)$.

Step 3: Update $u \leftarrow u + \Delta u$ and $\lambda \leftarrow \lambda + \Delta \lambda$ until the Frobenius norm of the right-hand side of Eq. (6.13) becomes smaller than a prescribed tolerance.

Note that, when (u, λ) is sufficiently close to the solution, the map E_{new} is nearly equal to E and the right coefficient of $\Delta\lambda$ becomes close to unity. We therefore employ the simplified

Newton method of the following form [17]:²⁰

$$E \Delta u + \Delta \lambda = B, \quad (6.14)$$

where E is the original $\mathbb{R}$ -linear map, $E : T_{U_0}\Sigma_0 \rightarrow T_U\Sigma$, and the right-hand side B is given by²¹

$$B = e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U U_{\text{new}}^{-1} - 1. \quad (6.15)$$

This equation can be efficiently solved using the projection introduced in Sect. 6.1. In fact, from the orthogonal decomposition

$$B = B_v + B_n = E B_{0,v} + F B_{0,n} \quad (6.16)$$

with $B_v \in T_U\Sigma$, $B_n \in N_U\Sigma$, and $B_{0,v} \in T_{U_0}\Sigma_0$ and $B_{0,n} \in N_{U_0}\Sigma_0$, we write

$$B = E B_{0,v} + B_n. \quad (6.17)$$

Comparing this with the left-hand side of the simplified Newton equation (6.14), we obtain the update

$$\Delta u = B_{0,v}, \quad \Delta \lambda = B_n. \quad (6.18)$$

A RATTLE step from (U, π) to (U', π') in $T\Sigma$ is summarized in Algorithm 2.

6.3. Summary of GT-HMC

We summarize the GT-HMC algorithm in Algorithm 3, that uses the simplified RATTLE integrator of Algorithm 2.

7. Algorithm of WV-HMC for group manifolds

In the WV-HMC algorithm, we consider the submanifold $\mathcal{S} = \mathcal{R}$, which represents the worldvolume of interest. The presentation below is structured in parallel with the previous section, in order to emphasize both the similarities and the differences between GT-HMC and WV-HMC.

²⁰This is the GT-HMC version of the fixed-point method introduced in Ref. [4], where HMC updates are considered for configurations on a (single connected component of) Lefschetz thimble.

²¹ B is not necessary traceless even when $G = SU(n)$ although the solution (u, λ) must be traceless for that case. A simple algorithm is that (1) we solve Eq. (6.14) with respect to Δu and $\Delta \lambda$ without imposing the traceless condition, (2) we make them traceless, and (3) we update u and λ as $u \leftarrow u + \Delta u$ and $\lambda \leftarrow \lambda + \Delta \lambda$. Explicit numerical tests in Sect. 8 show that the repetition of processes (1)–(3) converges quickly for the one-site model.

Algorithm 2 RATTLE $(U, \pi) \rightarrow (U', \pi')$ in GT-HMC with $U = U(t, U_0)$ and $\pi \in T_U \Sigma$

- 1: Compute $\xi(U) = [DS(U)]^\dagger$ and $Z = \epsilon\pi - (\epsilon^2/2)\xi(U)$
 - 2: Set $u = 0$ and $\lambda = 0$
 - 3: **for** $k = 0, 1, \dots$ **do**
 - 4: Compute $U_{\text{new}} = U(t, e^u U_0)$ and B [Eq. (6.15)]
 - 5: **if** $\|B\|$ is sufficiently small **then**
 - 6: **break**
 - 7: **end if**
 - 8: Decompose $B = EB_{0,v} + B_n$ by using Algorithm 1
 - 9: Update: $u \leftarrow u + B_{0,v}$, $\lambda \leftarrow \lambda + B_n$
 - 10: **end for**
 - 11: Set $U' = U_{\text{new}}$ and $\tilde{\pi}' = e^{(Z-\lambda)-(Z-\lambda)^\dagger} ((Z-\lambda)/\epsilon) e^{-(Z-\lambda)+(Z-\lambda)^\dagger} - (\epsilon/2)\xi(U')$
 - 12: Decompose $\tilde{\pi}' \in T_{U'} G^\mathbb{C}$ as $\tilde{\pi}' = \tilde{\pi}'_v + \tilde{\pi}'_n$ by using Algorithm 1, and set $\pi' = \tilde{\pi}'_v$
-

Algorithm 3 GT-HMC

- 1: Given $U \in \Sigma$, generate $\tilde{\pi} \in T_U G^\mathbb{C}$ from the Gaussian distribution $\propto e^{-(1/2)\langle \tilde{\pi}, \tilde{\pi} \rangle}$
 - 2: Project $\tilde{\pi}$ onto $T_U \Sigma$ by decomposing $\tilde{\pi} = \tilde{\pi}_v + \tilde{\pi}_n$, and set $\pi = \tilde{\pi}_v$
 - 3: Apply a sequence of MD steps using Algorithm 2 to evolve $(U, \pi) \rightarrow (U', \pi')$
 - 4: Compute $\Delta H \equiv H(U', \pi') - H(U, \pi)$ and accept U' as the new configuration with probability $\min(1, e^{-\Delta H})$. Otherwise, retain U as the next configuration
-

7.1. Projector in WV-HMC

We first describe an algorithm to project a vector $w \in T_U G^\mathbb{C}$ onto the tangent space $T_U \mathcal{R}$ and the normal space $N_U \mathcal{R}$:

$$w = w_\parallel + w_\perp \quad (w_\parallel = \Pi_{\mathcal{R}} w \in T_U \mathcal{R}, \quad w_\perp = (1 - \Pi_{\mathcal{R}})w \in N_U \mathcal{R}). \quad (7.1)$$

This decomposition can be carried out by making use of the projection onto the surface $\Sigma = \Sigma_t$ [see Eq. (6.1) and Algorithm 1]. In fact, let us decompose the vectors w and ξ as

$$w = w_v + w_n \quad (w_v \in T_U \Sigma_t, \quad w_n \in N_U \Sigma_t), \quad (7.2)$$

$$\xi = \xi_v + \xi_n \quad (\xi_v \in T_U \Sigma_t, \quad \xi_n \in N_U \Sigma_t). \quad (7.3)$$

Then, the components $w_\parallel$ and $w_\perp$ are given by²²

$$w_\parallel = w_v + c\xi_n, \quad w_\perp = w_n - c\xi_n \quad (7.4)$$

with

$$c = \frac{\langle \xi_n, w_n \rangle}{\langle \xi_n, \xi_n \rangle}. \quad (7.5)$$

²²One can easily show that $\langle \xi_n, w_\perp \rangle = 0$ and $\langle w_\parallel, w_\perp \rangle = 0$, confirming orthogonality.

7.2. Constrained molecular dynamics on $\mathcal{R}$

We begin by recalling the quantity we aim to evaluate [Eq. (3.41)]:

$$\langle \mathcal{O} \rangle = \frac{\langle \mathcal{F}(U) \mathcal{O}(U) \rangle_{\mathcal{R}}}{\langle \mathcal{F}(U) \rangle_{\mathcal{R}}}. \quad (7.6)$$

Here, the reweighted average over $\mathcal{R}$ is defined by [Eq. (3.57)]:

$$\langle g(U) \rangle_{\mathcal{R}} = \frac{\int_{T\mathcal{R}} d\Omega_{\mathcal{R}} e^{-H(U,\pi)} g(U)}{\int_{T\mathcal{R}} d\Omega_{\mathcal{R}} e^{-H(U,\pi)}}, \quad (7.7)$$

where [Eqs. (3.50), (3.56), (3.58), (3.59)]

$$d\Omega_{\mathcal{R}} = \frac{\omega_{\mathcal{R}}^{N+1}}{(N+1)!} \quad \text{with} \quad \omega_{\mathcal{R}} = d\langle \pi, \theta_{\mathcal{R}} \rangle, \quad (7.8)$$

$$H(U, \pi) = \frac{1}{2} \langle \pi, \pi \rangle + V(U, U^\dagger) \quad \text{with} \quad V(U, U^\dagger) = \text{Re } S(U) + W(t(U, U^\dagger)). \quad (7.9)$$

The reweighting factor $\mathcal{F}(U)$ takes the form [Eq. (3.43)]:

$$\mathcal{F}(U) = \alpha^{-1} \frac{\det E}{\sqrt{\gamma}} e^{-i \text{Im } S(U)}. \quad (7.10)$$

A single RATTLE step is given by

$$Z = \epsilon \pi - \epsilon^2 [DV(U, U^\dagger)]^\dagger, \quad (7.11)$$

$$U' = e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U, \quad (7.12)$$

$$\tilde{\pi}' = e^{(Z-\lambda)-(Z-\lambda)^\dagger} \frac{1}{\epsilon} (Z - \lambda) e^{-(Z-\lambda)+(Z-\lambda)^\dagger} - \epsilon [DV(U', U'^\dagger)]^\dagger, \quad (7.13)$$

$$\pi' = \Pi'_{\mathcal{R}} \tilde{\pi}'. \quad (7.14)$$

Here, the Lagrange multiplier $\lambda \in N_U \mathcal{R}$ is determined such that $U' \in \mathcal{R}$, which is equivalent to finding a triplet $(h \in \mathbb{R}, u \in T_{U_0} \Sigma_0, \lambda \in N_U \mathcal{R})$ satisfying the following equation for a given $U = U(t, U_0)$ (see Fig. 7):

$$e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U(t, U_0) = U(t + h, e^u U_0). \quad (7.15)$$

The gradient of the potential, $[DV(U, U^\dagger)]^\dagger$, can be set to the following form:

$$[DV(U, U^\dagger)]^\dagger = \frac{1}{2} \left[\xi + \frac{W'(t)}{\langle \xi_n, \xi_n \rangle} \xi_n \right]. \quad (7.16)$$

A proof is given in Appendix E.

We solve Eq. (7.15) using Newton's method based on the ansatz that an update $h \leftarrow h + \Delta h$, $u \leftarrow u + \Delta u$ and $\lambda \leftarrow \lambda + \Delta \lambda$ can be effectively performed by solving the equation

$$e^{-\Delta \lambda} e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U(t, U_0) = U(t + h + \Delta h, e^{\Delta u} e^u U_0). \quad (7.17)$$

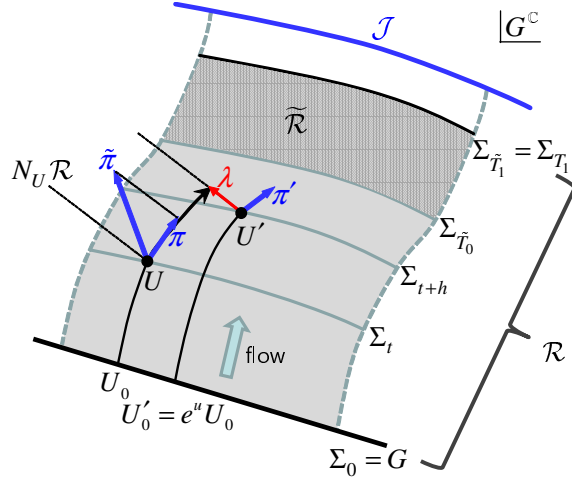


Figure 7: RATTLE in WV-HMC: $(U, \pi) \rightarrow (U', \pi')$. The initial momentum π is obtained by projecting a Gaussian distributed $\tilde{\pi} \in T_U G^{\mathbb{C}}$ onto the tangent space $T_U \mathcal{R}$. The original vector $\tilde{\pi}$ is sampled from $\propto e^{-(1/2)\langle \tilde{\pi}, \tilde{\pi} \rangle}$.

After the linearization, the algorithm proceeds as follows:

Step 1: Initialize $\Delta h = 0$, $\Delta u = 0$ and $\Delta \lambda = 0$.

Step 2: Solve the linearized equation

$$\xi_{\text{new}} \Delta h + E_{\text{new}} \Delta u + \Delta \lambda e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U U_{\text{new}}^{-1} = e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U U_{\text{new}}^{-1} - 1. \quad (7.18)$$

Here, $U = U(t, U_0)$, $U_{\text{new}} = U(t + h, e^u U_0)$, $\xi_{\text{new}} = \xi(U_{\text{new}})$, and E , E_{new} are the $\mathbb{R}$ -linear maps, $E : T_{U_0} \Sigma_0 \rightarrow T_U \Sigma_t$ and $E_{\text{new}} : T_{e^u U_0} \Sigma_0 \rightarrow T_{U_{\text{new}}} \Sigma_{t+h}$, respectively [see Eq. (3.11)].

Step 3: Update $h \leftarrow h + \Delta h$, $u \leftarrow u + \Delta u$, $\lambda \leftarrow \lambda + \Delta \lambda$ until the Frobenius norm of the right-hand side of Eq. (7.18) becomes smaller than a prescribed tolerance.

When (h, u, λ) is sufficiently close to the solution, we have $\xi_{\text{new}} \approx \xi$ and $E_{\text{new}} \approx E$, and the right coefficient of $\Delta \lambda$ is nearly unity. We therefore employ the simplified Newton method of the following form [17]:²³

$$\xi \Delta h + E \Delta u + \Delta \lambda = B, \quad (7.19)$$

where B is given by

$$B = e^{(Z-\lambda)-(Z-\lambda)^\dagger} e^{(Z-\lambda)^\dagger} U U_{\text{new}}^{-1} - 1. \quad (7.20)$$

This linear equation can be solved using the projection introduced in Sect. 6.1, which also applies to WV-HMC with minor modifications. To see this, we note that from the orthogonal

²³This is the WV-HMC version of the fixed-point method introduced in Ref. [4], where HMC updates are considered configurations on a (single connected component of) Lefschetz thimble.

decomposition

$$\xi = E\xi_{0,v} + \xi_n, \quad (7.21)$$

the left-hand side of Eq. (7.19) can be rewritten as

$$[E(\xi_{0,v} \Delta h + \Delta u) + \xi_n \Delta h] + \Delta \lambda. \quad (7.22)$$

Comparing this with the decomposition

$$B = B_{\parallel} + B_{\perp} = [EB_{0,v} + c_B \xi_n] + [B_n - c_B \xi_n] \quad \left(c_B \equiv \frac{\langle B, \xi_n \rangle}{\langle \xi_n, \xi_n \rangle} \right), \quad (7.23)$$

we obtain the following equations:

$$\xi_{0,v} \Delta h + \Delta u = B_{0,v}, \quad (7.24)$$

$$\Delta h = c_B, \quad (7.25)$$

$$\Delta \lambda = B_n - c_B \xi_n, \quad (7.26)$$

which can be solved explicitly as

$$\Delta h = c_B, \quad (7.27)$$

$$\Delta u = B_{0,v} - c_B \xi_{0,v}, \quad (7.28)$$

$$\Delta \lambda = B_n - c_B \xi_n. \quad (7.29)$$

A RATTLE step from (U, π) to (U', π') in $T\mathcal{R}$ is summarized in Algorithm 4.

7.3. Treatment of the boundary

In order to handle a worldvolume with finite temporal extent, we require the configurations in molecular dynamics to remain within the region $T_0 \lesssim t \lesssim T_1$. This is achieved by introducing the function $W(t)$ of the following form (see Refs. [17, 19]):

$$W(t) = \begin{cases} -\gamma(t - T_0) + c_0 (e^{(t-T_0)^2/2d_0^2} - 1) & (t < T_0) \\ -\gamma(t - T_0) & (T_0 \leq t \leq T_1) \\ -\gamma(t - T_0) + c_1 (e^{(t-T_1)^2/2d_1^2} - 1) & (t > T_1). \end{cases} \quad (7.30)$$

This term introduces soft reflective boundaries at $t = T_0$ (lower boundary) and $t = T_1$ (upper boundary), with penetration depths d_0 and d_1 , respectively. The coefficients c_0 and c_1 control the wall heights at $t = T_0 - d_0$ and $t = T_1 + d_1$, with gradients $-\gamma - c_0/d_0$ and $-\gamma + c_1/d_1$, respectively. However, due to the finite MD step size $\Delta s = \epsilon$, some configurations may penetrate deeply into the wall regions, resulting in a strong repulsive force,

Algorithm 4 RATTLE $(U, \pi) \rightarrow (U', \pi')$ in WV-HMC with $U = U(t, U_0)$ and $\pi \in T_U \mathcal{R}$

- 1: Compute $\xi(U) = [DS(U)]^\dagger$ and $Z = \epsilon\pi - \epsilon^2 [DS(U)]^\dagger$
 - 2: Set $h = 0$, $u = 0$, and $\lambda = 0$
 - 3: **for** $k = 0, 1, \dots$ **do**
 - 4: Compute $U_{\text{new}} = U(t + h, e^u U_0)$ and B [Eq. (7.20)]
 - 5: **if** $\|B\|$ is sufficiently small **then**
 - 6: **break**
 - 7: **end if**
 - 8: Decompose $B = EB_{0,v} + B_n$ and $\xi = E\xi_{0,v} + \xi_n$
 - 9: Set $\Delta h = c_B$, $\Delta u = B_{0,v} - c_B \xi_{0,v}$, $\Delta \lambda = B_n - c_B \xi_n$ with $c_B = \langle B, \xi_n \rangle / \langle \xi_n, \xi_n \rangle$
 - 10: Update: $h \leftarrow h + \Delta h$, $u \leftarrow u + \Delta u$, $\lambda \leftarrow \lambda + \Delta \lambda$
 - 11: **end for**
 - 12: Set $U' = U_{\text{new}}$ and $\tilde{\pi}' = e^{(Z-\lambda)-(Z-\lambda)^\dagger} ((Z - \lambda)/\epsilon) e^{-(Z-\lambda)+(Z-\lambda)^\dagger} - \epsilon [DS(U')]^\dagger$
 - 13: Decompose $\tilde{\pi}' \in T_{U'} G^{\mathbb{C}}$ as $\tilde{\pi}' = \tilde{\pi}'_{\parallel} + \tilde{\pi}'_{\perp}$ and set $\pi' = \tilde{\pi}'_{\parallel}$
-

Algorithm 5 Molecular dynamics step $(U, \pi) \rightarrow (U', \pi')$ in WV-HMC with boundary

- 1: Given (U, π) with $U = U(t, U_0)$, compute a trial molecular dynamics step $(U, \pi) \rightarrow (\tilde{U}, \tilde{\pi})$ with $\tilde{U} = U(\tilde{t}, \tilde{U}_0)$ using Algorithm 4
 - 2: **if** $T_0 - d_0 < \tilde{t} < T_1 + d_1$ **then**
 - 3: Set $U' = \tilde{U}$ and $\pi' = \tilde{\pi}$
 - 4: **else**
 - 5: Set $U' = U$ and $\pi' = -\pi$
 - 6: **end if**
-

$-W'(t) [Dt(U, U^\dagger)]^\dagger$, which appears in the force term $-[DV(U, U^\dagger)]^\dagger$ and may significantly degrade numerical stability. To avoid this issue, while preserving (1) exact volume preservation, (2) exact reversibility, and (3) approximate energy conservation with the same order of precision, a simple and effective solution is to reverse the trajectory when a configuration over-penetrates the boundary (see [15]). The procedure is summarized in Algorithm 5.

7.4. Summary of WV-HMC

We summarize the WV-HMC algorithm in Algorithm 6, that uses the simplified RATTLE integrator of Algorithm 4.

Algorithm 6 WV-HMC

- 1: Given $U \in \mathcal{R}$, generate $\tilde{\pi} \in T_U G^{\mathbb{C}}$ from the Gaussian distribution $\propto e^{-(1/2)\langle \tilde{\pi}, \tilde{\pi} \rangle}$
 - 2: Project $\tilde{\pi}$ onto $T_U \mathcal{R}$ by decomposing $\tilde{\pi} = \tilde{\pi}_{\parallel} + \tilde{\pi}_{\perp}$, and set $\pi = \tilde{\pi}_{\parallel}$
 - 3: Apply a sequence of MD steps using Algorithms 4 and 5 to evolve $(U, \pi) \rightarrow (U', \pi')$
 - 4: Compute $\Delta H \equiv H(U', \pi') - H(U, \pi)$ and accept U' as the new configuration with probability $\min(1, e^{-\Delta H})$. Otherwise, retain U as the next configuration
-

8. Numerical tests: one-site model

We validate the correctness of the proposed algorithm by performing numerical tests on the one-site model.

8.1. The model

The one-site model for a compact group $G = SU(n)$ is defined by the action

$$S(U) \equiv -\frac{\beta}{2n} \operatorname{tr}(U + U^{-1}) = \beta e(U), \quad (8.1)$$

where $e(U)$ is the energy “density”:

$$e(U) \equiv -\frac{1}{2n} \operatorname{tr}(U + U^{-1}). \quad (8.2)$$

We set β to be purely imaginary so that the action becomes complex-valued. Note that in this case, the Boltzmann weight becomes a pure phase factor, whose modulus is constant. This makes the naive reweighting method inapplicable. A well-defined probability distribution is obtained only after the action acquires the nonvanishing (positive) real part along flow subject to the flow equation (3.3).

To make the algebraic structure more transparent, we generalize the model to the form²⁴

$$S(U) = -a \operatorname{tr} U - b \operatorname{tr} U^{-1}, \quad (8.3)$$

where the original model corresponds to $a = b = \beta/(2n)$. We perform two validation checks to verify the correctness of the formalism. The first is to confirm that ΔH scales as ϵ^3 for a single MD step of step size $\Delta s = \epsilon$. This serves as a test of the algorithmic correctness of HMC. The second is to verify that analytical results are correctly reproduced. This checks the correctness of the constructed observables, including the reweighting factor $\mathcal{F}(U)$.

²⁴A topological term can be introduced by setting $a = \beta/(2n) + \theta/(4\pi)$ and $b = \beta/(2n) - \theta/(4\pi)$, which renders the action complex-valued even when both β and θ are real. Introducing a nonvanishing θ is meaningful when G includes a $U(1)$ factor.

In the following numerical examples, we consider the energy density $e(U)$ as the observable. The expectation value $\langle e \rangle$ for $G = SU(n)$ can be computed analytically from the partition function,

$$Z_G \equiv \int_G (dU) e^{a \operatorname{tr} U + b \operatorname{tr} U^{-1}} = \sum_{q \in \mathbb{Z}} (b/a)^{nq/2} \det[I_{q+j-i}(2\sqrt{ab})]_{i,j=1,\dots,n}, \quad (8.4)$$

where $I_k(z)$ denotes the modified Bessel function of the first kind. Then, the expectation value is given by $\langle e \rangle = -(\partial/\partial\beta) \ln Z_G$.

8.2. Flow equations

The variation of the action is given by ($\mathcal{P}$ denotes the traceless projector):

$$\begin{aligned} \delta S(U) &= -\operatorname{tr} [\delta U U^{-1} (aU - bU^{-1})] = -\operatorname{tr} [\delta U U^{-1} \mathcal{P}(aU - bU^{-1})] \\ &\equiv \operatorname{tr} [\delta U U^{-1} DS(U)]. \end{aligned} \quad (8.5)$$

This implies that

$$DS(U) = -\mathcal{P}(aU - bU^{-1}), \quad (8.6)$$

and therefore

$$\xi(U) = [DS(U)]^\dagger = -[\mathcal{P}(aU - bU^{-1})]^\dagger. \quad (8.7)$$

This defines the flow of a configuration:

$$\dot{U} = \xi(U) U \quad \text{with } U|_{t=0} = U_0. \quad (8.8)$$

For a tangent vector $v \in T_U \Sigma$,²⁵ we obtain

$$\begin{aligned} Hv &= D_v DS(U) = DS(e^v U)|_{v^1} = -\mathcal{P}(a e^v U - b U^{-1} e^{-v})|_{v^1} \\ &= -\mathcal{P}(avU + bU^{-1}v), \end{aligned} \quad (8.9)$$

which defines the flow of a tangent vector:

$$\dot{v} = (Hv)^\dagger + [\xi, v] \quad \text{with } v|_{t=0} = v_0. \quad (8.10)$$

For a normal vector $n \in N_U \Sigma$, we have²⁶

$$DD_n S(U) = D \operatorname{tr} [n DS(U)] = -D \operatorname{tr} [n(aU - bU^{-1})]. \quad (8.11)$$

²⁵The symbol $|_{v^1}$ indicates the linear part with respect to v .

²⁶The traceless projector $\mathcal{P}$ in $DS(U)$ can be omitted here, because n is traceless ($n \in N_U \Sigma \subset N_U G^{\mathbb{C}} \subset \operatorname{Lie} G^{\mathbb{C}}$).

This is evaluated from the variation,

$$\delta \operatorname{tr} [n(aU - bU^{-1})] = \operatorname{tr} [\delta U U^{-1}(aUn + bnU^{-1})] \equiv \operatorname{tr} [\delta U U^{-1} D \operatorname{tr} [n(aU - bU^{-1})]], \quad (8.12)$$

so that

$$H^T n = -D \operatorname{tr} n(aU - bU^{-1}) = -\mathcal{P}(aUn + bnU^{-1}), \quad (8.13)$$

which defines the flow of a normal vector:

$$\dot{n} = -(H^T n)^\dagger - [\xi^\dagger, n] \quad \text{with} \quad n|_{t=0} = n_0. \quad (8.14)$$

One can easily confirm that the identity $\langle n, v \rangle^\bullet = 0$ holds by using the explicit forms of the flow equations. We integrate the flow equations (8.8), (8.10) and (8.14) using an adaptive version of the Runge-Kutta-Munthe-Kaas algorithm [22, 23].

8.3. $G = SU(2)$ with purely imaginary coupling

In the $SU(2)$ case with $a = b = \beta/4$, the infinite series in Eq. (8.4) can be summed analytically, yielding $Z_{SU(2)} = (2/\beta) I_1(\beta)$, from which we obtain $\langle e \rangle = -I_2(\beta)/I_1(\beta)$. One can show that $\operatorname{Re} \langle e \rangle = 0$ for $SU(2)$. In fact, the partition function is an even function of β due to the Z_2 -invariance under $U \rightarrow -U$, so that $\langle e \rangle$ is an odd function of β , which implies that $\langle e \rangle$ must be purely imaginary when $\beta \in i\mathbb{R}$.

We set the parameters in $W(t)$ [Eq. (7.30)] to $T_0 = 0$, $T_1 = 0.5$, $\gamma = 0$, $c_0 = c_1 = 1$, $d_0 = d_1 = 0.1$. We expect that the solution (h, u, λ) to Eq. (7.19) scales with the MD step size $\Delta s = \epsilon$ as

$$h = O(\epsilon), \quad u = O(\epsilon), \quad \lambda = O(\epsilon^2). \quad (8.15)$$

Figure 8 exhibits $|h|$, $\|u\|$ and $\|\lambda\|$ at $\beta = i$ for various ϵ at flow time $t = 0.5$ with U_0 taken randomly, and confirms this expected scaling behavior. This scaling can be utilized to construct a good initial guess such that the simplified Newton method for solving Eq. (7.15) converges to the correct solution. A simple algorithm, e.g., is the following:

Step 1: Scale down $\epsilon \rightarrow \tilde{\epsilon} = f\epsilon$ with $f (\ll 1)$, to ensure that $\tilde{Z} \equiv Z|_{\epsilon \rightarrow \tilde{\epsilon}}$ in Eq. (5.42) is small, which in turn yields a solution close to the origin.

Step 2: Solve Eq. (7.19) with Z replaced by $\tilde{Z}$, obtaining $(\tilde{h}, \tilde{u}, \tilde{\lambda})$.

Step 3: Set the initial guess for the original equation to be $h_{\text{init}} \equiv \tilde{h}/f$, $u_{\text{init}} \equiv \tilde{u}/f$, $\lambda_{\text{init}} \equiv \tilde{\lambda}/f^2$.

The same trick also applies to GT-HMC.

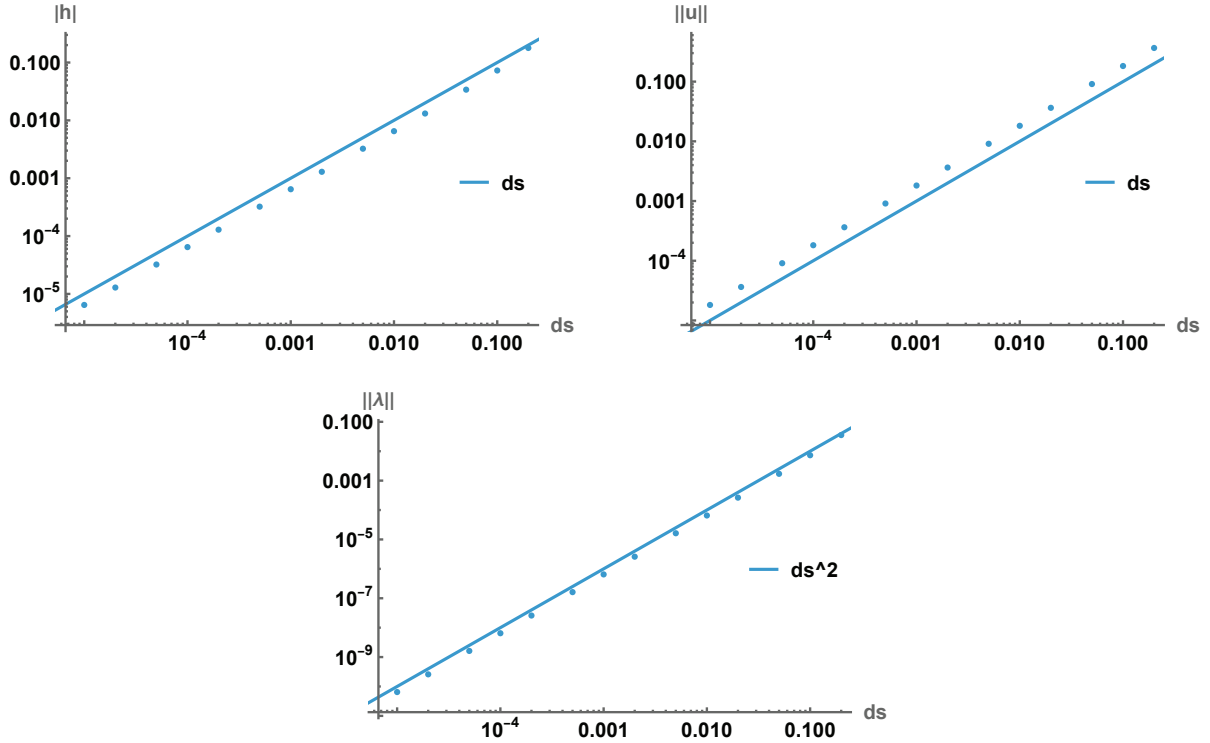


Figure 8: Scaling of (h, u, λ) as a function of ϵ in the $SU(2)$ one-site model with $\beta = i$. The Frobenius norm $||A||$ is used, defined as $||A|| \equiv (\text{tr } A^\dagger A)^{1/2}$.

Figure 9 shows that the deviation of energy, ΔH , of a single MD step scales as ϵ^3 , where the parameters are set as above, and molecular dynamics starts at flow time $t = 0.5$. This result strongly supports the algorithmic correctness of the proposed method.

In estimating the observable, we set the step size of molecular dynamics to $\Delta s = \epsilon = 0.01$ with $N_{\text{MD}} = 50$. We generate 2500 configurations using the WV-HMC algorithm in estimating $\langle e \rangle$. Figure 10 shows the real and imaginary parts of the energy density $\langle e \rangle$ for various values of $\beta \in i\mathbb{R}$. The results are in good agreement with the analytical values, supporting the correctness of the constructed observables, including the reweighting factor $\mathcal{F}(U)$.²⁷

8.4. $G = SU(3)$ with purely imaginary coupling

Other compact groups can be treated in the same way. Figure 11 shows plots of the energy density $\langle e \rangle$ for $SU(3)$ with various values of $\beta \in i\mathbb{R}$. The parameters are set to the same values as those of the $SU(2)$ case, except that the sample size is 2000. The numerical results are again in good agreement with the analytical results. This provides another strong

²⁷The observed large statistical errors in the real part can be suppressed if one introduces an algorithm which prompts the realization of Z_2 -symmetry $U \rightarrow -U$, which we have not employed in the present article.

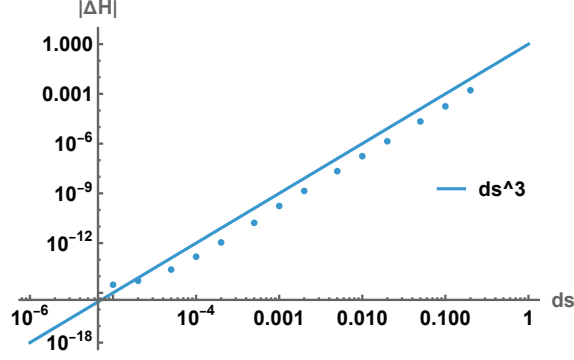


Figure 9: Deviation ΔH versus step sizes ϵ for the $SU(2)$ one-site model with $\beta = i$. Absolute tolerance and relative tolerance in recursive processes are set to 10^{-10} .

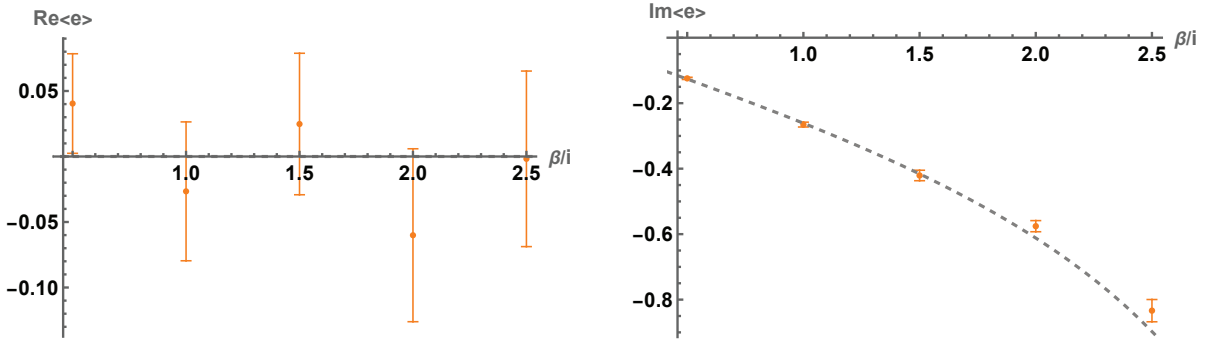


Figure 10: Real and imaginary parts of $\langle e \rangle$ in the $SU(2)$ one-site model for $\beta \in i\mathbb{R}$. Dashed lines show the analytic results.

evidence for the validity of the present formalism.

9. Conclusions and outlook

After developing a general theory of unconstrained and constrained Hamiltonian dynamics on $G^{\mathbb{C}}$, we have demonstrated that the WV-HMC (and also GT-HMC) algorithm can be extended to group manifolds. The key ingredient is to formulate the algorithm using a phase-space integral over the tangent bundle of the worldvolume, which naturally carries a symplectic structure. We have verified the correctness of the algorithms through numerical simulations of the one-site model.

A possible issue of the present formalism is the existence of the factor $\det E/\sqrt{\gamma}$ in the reweighting factor $\mathcal{F}(U)$ [see Eq. (3.43)]. This is not a pure phase factor generically, and thus has a chance to cause an overlap problem. Although we have not observed any problems in investigating the one-site model, this potential issue should be kept in mind when more complicated cases are considered.

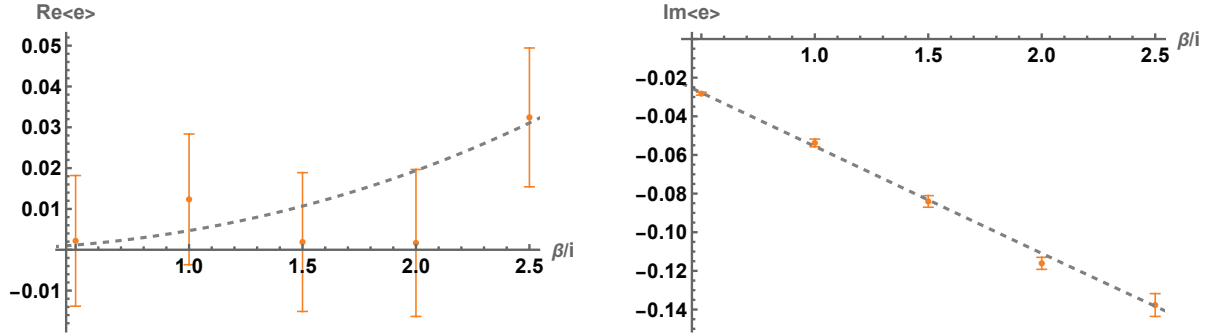


Figure 11: Real and imaginary parts of $\langle e \rangle$ in the $SU(3)$ one-site model for $\beta \in i\mathbb{R}$. Dashed lines show the analytic results.

The present formalism can be directly applied to lattice gauge theories without any modification to the algorithmic structure. The compact group G becomes a product group $G = \prod_{x,\mu} G_{x,\mu}$ [e.g., $G_{x,\mu} = SU(n)$ at each link (x, μ)], and the Lie algebra is accordingly given by $\text{Lie } G = \bigoplus_{x,\mu} \text{Lie } G_{x,\mu} = \bigoplus_{x,\mu,a} \mathbb{R} (T_{x,\mu})_a$ with commutation relations $[(T_{x,\mu})_a, (T_{y,\nu})_b] = \delta_{xy} \delta_{\mu\nu} C_{ab}^c (T_{x,\mu})_c$. A detailed study of lattice QCD with complex actions will be presented in subsequent publications.

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A. Hamiltonian dynamics on a symplectic manifold

In this appendix, we review Hamiltonian dynamics on a symplectic manifold, focusing on the material relevant to the main text.

A.1. Symplectic manifolds, symplectic forms and symplectic potentials

We begin by considering Hamiltonian dynamics on a general symplectic manifold $(\mathcal{M}, \omega)$. Here, $\mathcal{M} = \{x^i\}$ ($i = 1, \dots, 2N$) is a real $2N$ -dimensional manifold, and $\omega = (1/2) \omega_{ij}(x) dx^i \wedge$

dx^j is a *symplectic two-form* on $\mathcal{M}$ that satisfies the following conditions:

$$(1) \text{ Nondegeneracy: The matrix } (\omega_{ij}) \text{ is invertible, with inverse denoted by } (\omega^{ij}). \quad (\text{A.1})$$

$$(2) \text{ Closedness: The exterior derivative of } \omega \text{ vanishes, i.e., } d\omega = 0. \quad (\text{A.2})$$

Condition (1) allows us to define the Poisson bracket of smooth functions $f(x)$ and $g(x)$ on $\mathcal{M}$ by²⁸

$$\{f, g\}(x) \equiv \omega^{ij}(x) \partial_i f(x) \partial_j g(x) = -\{g, f\}(x). \quad (\text{A.3})$$

Condition (2) ensures that the Poisson bracket satisfies the Jacobi identity, $\{\{f, g\}, h\} = \{\{f, h\}, g\} + \{f, \{g, h\}\}$. Moreover, the closedness of ω implies the (local) existence of a one-form $a = a_i(x) dx^i$, called the *symplectic potential*, such that

$$\omega = da, \quad (\text{A.4})$$

which in components reads as $\omega_{ij} = \partial_i a_j - \partial_j a_i$.

A.2. Finite symplectic transformations and pull-back

Given a function $f(x)$, we define the associated vector field $v_f = v_f^i(x) \partial_i$ and the transformation T_f as

$$v_f^i(x) \equiv \omega^{ij}(x) \partial_j f(x). \quad (\text{A.5})$$

$$T_f \equiv e^{\{*, f\}}. \quad (\text{A.6})$$

We define the action of T_f on an n -form $A = (1/n!) A_{i_1 \dots i_n}(x) dx^{i_1} \wedge \dots \wedge dx^{i_n}$ as

$$T_f A \equiv \frac{1}{n!} (T_f A_{i_1 \dots i_n})(x) d(T_f x^{i_1}) \wedge \dots \wedge d(T_f x^{i_n}). \quad (\text{A.7})$$

The Poisson bracket $\{*, f\}$ acts on a function $G(x)$ as

$$\{G, f\} = \omega^{ij} \partial_i G \partial_j f = v_f^i \partial_i G = v_f G, \quad (\text{A.8})$$

which implies that

$$(T_f G)(x) = (e^{v_f} G)(x). \quad (\text{A.9})$$

This shows that the action of T_f on differential forms can be interpreted as the pull-back $(e^{v_f})^*$ of the diffeomorphism e^{v_f} ,

$$T_f A = (e^{v_f})^* A = e^{\mathcal{L}_{v_f}} A. \quad (\text{A.10})$$

²⁸In particular, we have $\{x^i, x^j\} = \omega^{ij}(x)$.

Here, $\mathcal{L}_v$ denotes the Lie derivative along a vector field $v = v^i(x) \partial_i$, which is given in terms of the exterior derivative $d = dx^i \wedge \partial/\partial_i$ and the interior product i_v by the Cartan formula:²⁹

$$\mathcal{L}_v = i_v d + d i_v. \quad (\text{A.11})$$

The transformation $T_f = e^{\{*,f\}} = e^{\mathcal{L}_{v_f}}$ is a symplectic transformation, meaning that

$$T_f \omega = \omega. \quad (\text{A.12})$$

To see this, note first that

$$\mathcal{L}_{v_f} a = (i_{v_f} d + d i_{v_f}) a = i_{v_f} \omega + d(i_{v_f} a). \quad (\text{A.13})$$

Since $i_{v_f} \omega$ can be rewritten as

$$i_{v_f} \omega = v_f^j \omega_{ji} dx^i = \omega^{jk} \partial_k f \omega_{ji} dx^i = -\partial_i f dx^i = -df, \quad (\text{A.14})$$

we have

$$\mathcal{L}_{v_f} a = d(i_{v_f} a - f). \quad (\text{A.15})$$

This implies that³⁰

$$\mathcal{L}_{v_f} \omega = d(\mathcal{L}_{v_f} a) = 0, \quad (\text{A.16})$$

thus confirming the invariance of ω under $T_f = e^{\mathcal{L}_{v_f}}$.

A.3. Hamiltonian dynamics: continuum version

We consider Hamiltonian dynamics on $\mathcal{M}$ with a Hamiltonian $H(x)$. A trajectory in $\mathcal{M}$ is described by a smooth function $x(s)$ of a continuous parameter s . We characterize such trajectories as those that minimize the first-order action

$$I[x(s)] \equiv \int ds [a_i(x) \overset{\circ}{x}^i - H(x)], \quad (\text{A.17})$$

where $\overset{\circ}{x}^i(s) \equiv (d/ds) x^i(s)$. The resulting equations of motion (Hamilton's equations) are given by

$$\overset{\circ}{x}^i = \omega^{ij}(x) \partial_j H(x) = \{x^i, H(x)\}. \quad (\text{A.18})$$

²⁹The interior product acts on an n -form as $i_v A = (1/(n-1)!) (i_v A)_{i_1 \dots i_{n-1}}(x) dx^{i_1} \wedge \dots \wedge dx^{i_{n-1}}$ with $(i_v A)_{i_1 \dots i_{n-1}} = v^i A_{i i_1 \dots i_{n-1}}$.

³⁰We have used the identity $\mathcal{L}_v(dA) = d(\mathcal{L}_v A)$, which follows from Eq. (A.11) and the nilpotency $d^2 = 0$. This identity also implies $T_f(dA) = d(T_f A)$ because $T_f = e^{\mathcal{L}_{v_f}}$.

One can show that both the symplectic form and the Hamiltonian are conserved along the trajectory:

$$\dot{\omega} = 0, \quad (\text{A.19})$$

$$\dot{H} = 0. \quad (\text{A.20})$$

Symplecticity [Eq.(A.19)] follows from the fact that the Hamiltonian flow is a symplectic transformation [replacing $f(x)$ in Eq. (A.16) with $H(x)$], $\dot{\omega} = \mathcal{L}_{v_H}\omega = 0$. The energy conservation [Eq. (A.20)] is shown as follows:

$$\dot{H} = \partial_i H \dot{x}^i = \omega^{ij} \partial_i H \partial_j H = 0. \quad (\text{A.21})$$

The finite-time evolution of x over an interval $\Delta s = \epsilon$ is given by

$$T_{\epsilon H} x = e^{\epsilon \{*, H\}} x. \quad (\text{A.22})$$

A.4. Hamiltonian dynamics: discrete version

We focus here on the case where $\mathcal{M}$ is equipped with canonical coordinates $x = (x^i) = (q^a, p_a)$ ($a = 1, \dots, N$) in which the symplectic potential and the symplectic form are given by $\theta = p_a dq^a$ and $\omega = da = dp_a \wedge dq^a$, respectively. We assume that the Hamiltonian takes the separable form,

$$H(q, p) = K(p) + V(q) \text{ with } K(p) \equiv \frac{1}{2} p_a^2. \quad (\text{A.23})$$

We discretize the continuum “transfer matrix” $T_H = e^{\epsilon \{*, H\}}$ with step size $\Delta s = \epsilon$ in the symmetric form,³¹

$$T \equiv T_{V/2} T_K T_{V/2}, \quad (\text{A.24})$$

whose deviation from T_H is $O(\epsilon^3)$,³²

$$T = T_H + O(\epsilon^3). \quad (\text{A.25})$$

The following identities are straightforward to verify:

$$T_{V/2} q = q, \quad (\text{A.26})$$

$$T_{V/2} p = p - \frac{\epsilon}{2} \partial V(q) \quad (\equiv p_{1/2}), \quad (\text{A.27})$$

$$T_K q = q + \epsilon p, \quad (\text{A.28})$$

$$T_K p = p, \quad (\text{A.29})$$

³¹For simplicity, we denote $e^{\epsilon \{*, f\}}$ by T_f (not $T_{\epsilon f}$).

³²This follows from the identity $e^{\epsilon A/2} e^{\epsilon B} e^{\epsilon A/2} = e^{\epsilon(A+B)} + O(\epsilon^3)$, which holds for linear operators A, B of any type.

from which the update rule $T : (q, p) \rightarrow (q', p')$ is given by:

$$\begin{aligned}
q' &\equiv Tq = T_{V/2} T_K T_{V/2} q \\
&= T_{V/2} T_K q \\
&= T_{V/2} (q + \epsilon p) \\
&= q + \epsilon p_{1/2},
\end{aligned} \tag{A.30}$$

$$\begin{aligned}
p' &\equiv Tp = T_{V/2} T_K T_{V/2} p \\
&= T_{V/2} T_K [p - (\epsilon/2) \partial V(q)] \\
&= T_{V/2} [p - (\epsilon/2) \partial V(q + \epsilon p)] \\
&= p_{1/2} - (\epsilon/2) \partial V(q + \epsilon p_{1/2}) \\
&= p_{1/2} - \frac{\epsilon}{2} \partial V(q').
\end{aligned} \tag{A.31}$$

In practice, this update is implemented as the following three steps:

$$p_{1/2} = p - \frac{\epsilon}{2} \partial V(q), \tag{A.32}$$

$$q' = q + \epsilon p_{1/2}, \tag{A.33}$$

$$p' = p_{1/2} - \frac{\epsilon}{2} \partial V(q'), \tag{A.34}$$

which form the basis of the standard leapfrog algorithm. When the above set is repeated N_{MD} times, the intermediate steps can be combined and the full trajectory is generated as:

$$\begin{aligned}
&p_{1/2} = p - \frac{\epsilon}{2} \partial V(q) \\
&q' = q + \epsilon p_{1/2} \\
&\textbf{for } i = 1, \dots, N_{\text{MD}} - 1 \textbf{ do} \\
&\quad p_{1/2} \leftarrow p_{1/2} - \epsilon \partial V(q') \\
&\quad q' \leftarrow q' + \epsilon p_{1/2} \\
&\textbf{end do} \\
&p' = p_{1/2} - \frac{\epsilon}{2} \partial V(q')
\end{aligned} \tag{A.35}$$

One can show that the leapfrog update (A.32)–(A.34) satisfies the following three key properties:

(1) Exact reversibility: The transformation is invariant under time reversal,

$$q \rightarrow \tilde{q} \equiv q', \tag{A.36}$$

$$p \rightarrow \tilde{p} \equiv -p', \tag{A.37}$$

$$p_{1/2} \rightarrow \tilde{p}_{1/2} \equiv -p_{1/2}, \tag{A.38}$$

$$q' \rightarrow \tilde{q}' \equiv q. \tag{A.39}$$

(2) Symplecticity: The symplectic form ω (and thus the volume form $d\Omega \equiv \omega^N/N! = dq dp$) is preserved,

$$\omega' = \omega. \quad (\text{A.40})$$

(3) Approximate energy condition:

$$H(q', p') = H(q, p) + O(\epsilon^3). \quad (\text{A.41})$$

Property (1) is obvious. Property (2) should also be obvious, because the transformation T is a composition of symplectic transformations [Eq. (A.24)]. It can also be shown explicitly by using

$$T_{V/2} a = T_{V/2} (p dq) = a - \frac{\epsilon}{2} dV(q), \quad (\text{A.42})$$

$$T_K a = T_K (p dq) = a + \frac{\epsilon}{2} d(p^2), \quad (\text{A.43})$$

from which it follows that

$$T a = a + \epsilon d \left\{ \frac{1}{2} p_{1/2}^2 - \frac{1}{2} [V(q) + V(q')] \right\}, \quad (\text{A.44})$$

and thus

$$T\omega = Tda = d(Ta) = da = \omega. \quad (\text{A.45})$$

Property (3) follows from the identity $T = T_H + O(\epsilon^3)$ [Eq. (A.25)] and the fact that the exact flow conserves energy, $T_H H = e^{\{*, H\}} H = H$:

$$\begin{aligned} H(q', p') &= (TH)(q, p) = (T_H H)(q, p) + O(\epsilon^3) \\ &= H(q, p) + O(\epsilon^3). \end{aligned} \quad (\text{A.46})$$

A.5. Hybrid Monte Carlo (HMC)

We now consider the numerical evaluation of a path integral of the form

$$\langle \mathcal{O}(x) \rangle = \frac{\int_{\mathbb{R}^N} dq e^{-V(q)} \mathcal{O}(q)}{\int_{\mathbb{R}^N} dq e^{-V(q)}}, \quad (\text{A.47})$$

which can be rewritten as an integral over the phase space $T\mathbb{R}^N \equiv \{(q, p)\}$,³³

$$\langle \mathcal{O}(q) \rangle = \frac{\int_{T\mathbb{R}^N} d\Omega e^{-H(q, p)} \mathcal{O}(q)}{\int_{T\mathbb{R}^N} d\Omega e^{-H(q, p)}}. \quad (\text{A.48})$$

³³We regard the tangent bundle (rather than cotangent bundle) as the phase bundle. This picture is generally preferable when discussing molecular dynamics.

Here, $d\Omega = \omega^N/N! = dq dp$ is the symplectic volume element and $H(q, p) = (1/2)p_a^2 + V(q)$ is the Hamiltonian. This expectation value can be estimated by a sample average over configurations $\{(q, p)\}$ drawn from the distribution $\propto e^{-H(q, p)}$.

To construct a Markov chain whose equilibrium distribution is $\propto e^{-H(q, p)}$ with the Hamiltonian satisfying $H(q, -p) = H(q, p)$ (as in the present case), we introduce a set of stochastic processes $P_{(k)}(q', p' | q, p)$ ($k = 1, \dots, K$) such that each satisfies the detailed balance condition of molecular dynamics,

$$P_{(k)}(q', p' | q, p) e^{-H(q, p)} = P_{(k)}(q, -p | q', -p') e^{-H(q', -p')}, \quad (\text{A.49})$$

and that their product $P \equiv P_{(K)} \cdots P_{(1)}$ possesses a proper ergodicity property that ensures convergence to a unique equilibrium. One can easily verify that $e^{-H(q, p)}$ is the (unique) eigenvector of $P(q', p' | q, p)$ of unit eigenvalue,

$$\int_{T\mathbb{R}^N} d\Omega' P(q, p | q', p') e^{-H(q', p')} = e^{-H(q, p)}, \quad (\text{A.50})$$

implying that $e^{-H(q, p)} / \int_{T\mathbb{R}^N} d\Omega e^{-H(q, p)}$ is the stationary distribution reached by the Markov chain defined by P .

In the present case, we employ the following two stochastic processes:

(1) Heat bath for p :

$$P_{(1)}(q', p' | q, p) = e^{-(1/2)p_a'^2} \delta(q' - q). \quad (\text{A.51})$$

(2) Molecular dynamics (MD) plus Metropolis test:

$$\begin{aligned} P_{(2)}(q', p' | q, p) \\ = \min(1, e^{-[H(q', p') - H(q, p)]}) \delta_{T\mathbb{R}^N}((q', p') - T^{N_{\text{MD}}}(q, p)) \quad \text{for } (q', p') \neq (q, p). \end{aligned} \quad (\text{A.52})$$

Here, T denotes a single MD update (i.e., the composition of Eqs. (A.32)–(A.34)), and $T^{N_{\text{MD}}}$ denotes its repetition N_{MD} times [cf. Eq. (A.35)]. $\delta_{T\mathbb{R}^N}(q, p) = \delta(q) \delta(p)$ is the delta function with respect to the volume form $d\Omega$. The volume preservation ensures that the Jacobian of the map $(q, p) \mapsto (q', p') = T^{N_{\text{MD}}}(q, p)$ is unity. The reversibility together with the unit Jacobian then shows

$$\delta_{T\mathbb{R}^N}((q', p') - T^{N_{\text{MD}}}(q, p)) = \delta_{T\mathbb{R}^N}((q, -p) - T^{N_{\text{MD}}}(q', -p')), \quad (\text{A.53})$$

from which one can show that $P_{(2)}$ satisfies the detailed balance condition (A.49). Although neither $P_{(1)}$ nor $P_{(2)}$ is ergodic, their product $P \equiv P_{(2)}P_{(1)}$ is typically ergodic, guaranteeing convergence to the correct equilibrium.

When the observables depend only on q , the combined algorithm $P = P_{(2)}P_{(1)}$ can be interpreted as a stochastic process on q as follows [18]:

- Step 1 (momentum refresh):
Given $q \in \mathbb{R}^N$, generate $p = (p_a)$ from the Gaussian distribution $\propto e^{-(1/2)p_a^2}$.
- Step 2 (molecular dynamics):
Evolve $(q, p) \rightarrow (q', p')$ using the MD algorithm [Eq. (A.35)].
- Step 3 (Metropolis test):
Accept the proposed q' with probability $\min(1, e^{-[H(q', p') - H(q, p)]})$.

B. HMC algorithm for a compact group

In this appendix, we review the Hybrid Monte Carlo (HMC) algorithm for a compact Lie group G of dimension N [18]. Although the material presented here is likely familiar to experts, we present it in a form that parallels the structure of the main text. In this appendix only, elements of G are denoted by the symbol U (rather than U_0 , as used in the main text). We assume that each group element is uniquely represented by a unitary matrix. Thus, elements of the Lie algebra $\text{Lie } G$ are anti-hermitian matrices. We choose a basis $\{T_a\}$ ($a = 1, \dots, N$) of $\text{Lie } G$ such that $\text{tr } T_a T_b = -\delta_{ab}$. Accordingly, we raise and lower indices using the rule $A^a = -A_a$. Throughout this appendix, we employ the concepts and technical terms introduced in Appendix A.

B.1. Hamiltonian dynamics on a compact group

Our phase space is the tangent bundle over G ,

$$TG = \{(U, \pi) \mid U = (U_{ij}) \in G, \pi = (\pi_{ij}) = T_a \pi^a \in T_U G\}. \quad (\text{B.1})$$

To *define* consistent Hamiltonian dynamics on TG with a given Hamiltonian of the form

$$H(U, \pi) = \frac{1}{2} \text{tr } \pi^\dagger \pi + V(U) = -\frac{1}{2} \text{tr } \pi^2 + V(U), \quad (\text{B.2})$$

we consider the first-order action

$$I[U(s), \pi(s)] = \int ds [\text{tr } \pi^\dagger \overset{\circ}{U} U^{-1} - H(U, \pi)], \quad (\text{B.3})$$

where $\overset{\circ}{U} \equiv dU/ds$. Hamilton's equations are obtained by varying $I[U, \pi]$, and are given by

$$\overset{\circ}{U} = \pi U, \quad (\text{B.4})$$

$$\overset{\circ}{\pi} = DV(U) + [\overset{\circ}{U} U^{-1}, \pi] = DV(U). \quad (\text{B.5})$$

In components, these are expressed as

$$\overset{\circ}{U}_{ij} = (\pi U)_{ij}, \quad (B.6)$$

$$\overset{\circ}{\pi}^a = D^a V(U) = -D_a V(U). \quad (B.7)$$

B.2. Symplectic structure of TG

The first-order action determines the symplectic potential as

$$a = \text{tr } \pi^\dagger \theta = -\text{tr } \pi \theta. \quad (B.8)$$

This gives the symplectic form as

$$\begin{aligned} \omega &= da \\ &= -\text{tr} (d\pi \theta + \pi \theta \wedge \theta) \\ &= \frac{1}{2} (dU_{ij} \ d\pi_{ij}) \begin{pmatrix} a_{ij,kl} & \delta_{li} U_{jk}^{-1} \\ -\delta_{jk} U_{li}^{-1} & 0 \end{pmatrix} \begin{pmatrix} dU_{kl} \\ d\pi_{kl} \end{pmatrix} \end{aligned} \quad (B.9)$$

with

$$a_{ij,kl} = -U_{jk}^{-1} (U^{-1} \pi)_{li} + U_{li}^{-1} (U^{-1} \pi)_{jk}. \quad (B.10)$$

The 2×2 block matrix can be inverted analytically, yielding the following Poisson brackets for $G = U(n)$ (or $G = SU(n)$):

$$\{U_{ij}, U_{kl}\} = 0, \quad (B.11)$$

$$\{U_{ij}, \pi_{kl}\} = -\delta_{li} U_{jk}^{-1} \left(+\frac{1}{n} U_{ij} \delta_{kl} \right), \quad (B.12)$$

$$\{\pi_{ij}, U_{kl}\} = \delta_{jk} U_{li}^{-1} \left(-\frac{1}{n} U_{ij} \delta_{kl} \right), \quad (B.13)$$

$$\{\pi_{ij}, \pi_{kl}\} = -\delta_{il} \pi_{kj} + \delta_{kj} \pi_{il}. \quad (B.14)$$

To derive these Poisson brackets in such a manner that manifestly preserves the traceless condition of π , it is preferable to express π in terms of its Lie algebra components π_a . Then, the symplectic potential becomes

$$a = -\pi_a \theta^a, \quad (B.15)$$

and the symplectic form $\omega = da$ is given by

$$\begin{aligned} \omega &= -d\pi_a \wedge \theta^a - \frac{1}{2} C_{ab}{}^c \pi_c \wedge \theta^a \\ &= \frac{1}{2} (\theta^a \ d\pi_a) \begin{pmatrix} -C_{ab}{}^c \pi_c & \delta_a^b \\ -\delta_b^a & 0 \end{pmatrix} \begin{pmatrix} \theta^b \\ d\pi_b \end{pmatrix}. \end{aligned} \quad (B.16)$$

The 2×2 block matrix can be easily inverted to give

$$\{U_{ij}, U_{kl}\} = 0, \quad (\text{B.17})$$

$$(\{U, \pi_b\} U^{-1})^a = -\delta_b^a \Leftrightarrow \{U_{ij}, \pi_a\} = -(T_a U)_{ij}, \quad (\text{B.18})$$

$$(\{\pi_a, U\} U^{-1})^b = \delta_a^b \Leftrightarrow \{\pi_a, U_{ij}\} = (T_a U)_{ij}, \quad (\text{B.19})$$

$$\{\pi_a, \pi_b\} = -C_{ab}^c \pi_c. \quad (\text{B.20})$$

Contracting both sides with T_{kl}^a and using the identity

$$T_{a,ij} T_{kl}^a = \delta_{il} \delta_{jk} \left(-\frac{1}{n} \delta_{ij} \delta_{kl} \right), \quad (\text{B.21})$$

we recover the Poisson brackets (B.11)–(B.14).

Using the Poisson brackets, one obtains the following identities:

$$\{U, K\} = \pi U, \quad \{U, V\} = 0, \quad (\text{B.22})$$

$$\{\pi, K\} = 0, \quad \{\pi, V\} = DV, \quad (\text{B.23})$$

which reproduce Hamilton's equations (B.4) and (B.5) in the form $\dot{U} = \{U, H\}$ and $\dot{\pi} = \{\pi, H\}$.

B.3. Leapfrog algorithm

For a function $f = f(U, \pi)$, we introduce the transformation T_f as

$$T_f \equiv e^{\epsilon \{*, f\}}, \quad (\text{B.24})$$

and define the transfer matrix of step size $\Delta s = \epsilon$ with respect to the Hamiltonian $H(U, \pi) = K(\pi) + V(U)$ by

$$T \equiv T_{V/2} T_K T_{V/2} = T_H + O(\epsilon^3). \quad (\text{B.25})$$

Expanding the exponentials, we find

$$T_{V/2} U = U, \quad (\text{B.26})$$

$$T_{V/2} \pi = \pi - \frac{\epsilon}{2} DV(U) \quad (\equiv \pi_{1/2}), \quad (\text{B.27})$$

$$T_K U = e^{\epsilon \pi} U, \quad (\text{B.28})$$

$$T_K \pi = \pi. \quad (\text{B.29})$$

It follows that

$$\begin{aligned} U' &\equiv TU = T_{V/2} T_K T_{V/2} U = T_{V/2} e^{\epsilon\pi} U \\ &= e^{\epsilon\pi_{1/2}} U, \end{aligned} \tag{B.30}$$

$$\begin{aligned} \pi' &\equiv T\pi = T_{V/2} T_K T_{V/2} \pi = T_{V/2} T_K \left[\pi - \frac{\epsilon}{2} DV(U) \right] \\ &= T_{V/2} \left[\pi - \frac{\epsilon}{2} DV(e^{\epsilon\pi} U) \right] = \pi_{1/2} - \frac{\epsilon}{2} DV(e^{\epsilon\pi_{1/2}} U) \\ &= \pi_{1/2} - \frac{\epsilon}{2} DV(U'), \end{aligned} \tag{B.31}$$

which leads to the following leapfrog update:

$$\pi_{1/2} = \pi - \frac{\epsilon}{2} DV(U), \tag{B.32}$$

$$U' = e^{\epsilon\pi_{1/2}} U, \tag{B.33}$$

$$\pi = \pi_{1/2} - \frac{\epsilon}{2} DV(U'). \tag{B.34}$$

Repeating the update N_{MD} times, the full leapfrog integrator becomes:

$$\begin{aligned} \pi_{1/2} &= \pi - \frac{\epsilon}{2} DV(U) \\ U' &= e^{\epsilon\pi_{1/2}} U \\ \textbf{for } i &= 1, \dots, N_{\text{MD}} - 1 \textbf{ do} \\ &\quad \pi_{1/2} \leftarrow \pi_{1/2} - \epsilon DV(U') \\ &\quad U' \leftarrow e^{\epsilon\pi_{1/2}} U' \\ \textbf{end do} \\ \pi' &= \pi_{1/2} - \frac{\epsilon}{2} DV(U') \end{aligned} \tag{B.35}$$

The leapfrog update (B.32)–(B.34) satisfies the following three properties:

(1) Exact reversibility: The transformation is invariant under time reversal,

$$U \rightarrow \tilde{U} \equiv U', \tag{B.36}$$

$$\pi \rightarrow \tilde{\pi} \equiv -\pi', \tag{B.37}$$

$$\pi_{1/2} \rightarrow \tilde{\pi}_{1/2} \equiv -\pi_{1/2}, \tag{B.38}$$

$$\pi' \rightarrow \tilde{\pi}' \equiv \pi. \tag{B.39}$$

(2) Symplecticity: The symplectic form ω (and thus the volume form $d\Omega = \omega^N/N!$) is preserved:

$$\omega' = \omega. \tag{B.40}$$

(3) Approximate energy conservation:

$$H(U', \pi') = H(U, \pi) + O(\epsilon^3). \quad (\text{B.41})$$

Properties (1) and (3) can be shown in a similar way to the flat case. Property (2) immediately follows from the fact that T is a composition of symplectic transformations, but can be shown explicitly by noting that

$$T_{V/2} \theta = \theta, \quad (\text{B.42})$$

$$T_K \theta = (de^{\epsilon\pi}) e^{-\epsilon\pi} + e^{\epsilon\pi} \theta e^{-\epsilon\pi}, \quad (\text{B.43})$$

and thus that

$$\begin{aligned} T_{V/2} a &= -\text{tr} (T_{V/2} \pi) (T_{V/2} \theta) = -\text{tr} \left[\pi - \frac{\epsilon}{2} DV(U) \right] \theta \\ &= a + \frac{\epsilon}{2} d\Omega(U), \end{aligned} \quad (\text{B.44})$$

$$\begin{aligned} T_K a &= -\text{tr} (T_K \pi) (T_K \theta) = -\text{tr} \pi [(de^{\epsilon\pi}) e^{-\epsilon\pi} + e^{\epsilon\pi} \theta e^{-\epsilon\pi}] = a - \text{tr} \pi (de^{\epsilon\pi}) e^{-\epsilon\pi} \\ &= a - \frac{\epsilon}{2} d[\text{tr} \pi^2], \end{aligned} \quad (\text{B.45})$$

which leads to

$$T_{V/2} \omega = \omega, \quad T_K \omega = \omega, \quad (\text{B.46})$$

and thus

$$T\omega = \omega. \quad (\text{B.47})$$

B.4. Hybrid Monte Carlo (HMC)

We aim to compute the expectation values of observables $\mathcal{O}(U)$ under the potential $V(U)$:

$$\langle \mathcal{O} \rangle = \frac{\int_G (dU) e^{-V(U)} \mathcal{O}(U)}{\int_G (dU) e^{-V(U)}}. \quad (\text{B.48})$$

Here, (dU) denotes the Haar measure on G defined using the Maurer-Cartan form $\theta = dUU^{-1} = T_a \theta^a$ as [see Eq. (2.4)]:

$$(dU) = \theta^1 \wedge \cdots \wedge \theta^N \quad (N = \dim G). \quad (\text{B.49})$$

The symplectic structure ω is derived from the symplectic potential $a = -\text{tr} \pi \theta = -\pi_a \theta^a$ as

$$\omega = da = -\text{tr} [d\pi \wedge \theta + \pi \theta \wedge \theta] = -d\pi_a \wedge \theta^a - \frac{1}{2} C_{ab}^c \pi_c \theta^a \wedge \theta^b, \quad (\text{B.50})$$

and the volume form $d\Omega$ of TG can be written as follows:

$$\begin{aligned}
d\Omega &\equiv \frac{1}{N!} \omega^N \\
&= \frac{1}{N!} (\langle d\pi, \theta \rangle + \langle \pi, \theta \wedge \theta \rangle)^N \\
&= \frac{1}{N!} \langle d\pi, \theta \rangle^N \\
&= \left(\prod_a d\pi_a \right) (dU).
\end{aligned} \tag{B.51}$$

Note that terms involving $\theta \wedge \theta$ vanish when taking the N -th wedge product ω^N . Thus, the expectation value (B.48) can be written as an integral over TG :

$$\langle \mathcal{O} \rangle = \frac{\int_{TG} d\Omega e^{-H(U, \pi)} \mathcal{O}(U)}{\int_{TG} d\Omega e^{-H(U, \pi)}} \tag{B.52}$$

with $H(U, \pi) = (1/2) \text{tr} \pi^\dagger \pi + V(U) = (1/2) (\pi_a)^2 + V(U)$.

As discussed in Sect. A.5, we construct the target distribution $\propto e^{-H(U, \pi)}$ through the following two stochastic processes:

(1) Heat bath for π :

$$P_{(1)}(U', \pi' | U, \pi) = e^{-(1/2) \text{tr} \pi'^\dagger \pi'} \delta(U', U), \tag{B.53}$$

where $\delta(U', U)$ is the bi-invariant delta function with respect to the Haar measure on G .

(2) Molecular dynamics (MD) plus Metropolis test:

$$\begin{aligned}
P_{(2)}(U', \pi' | U, \pi) \\
= \min(1, e^{-[H(U', \pi') - H(U, \pi)]}) \delta_{TG}((U', \pi'), T^{N_{\text{MD}}}(U, \pi)) \text{ for } (U', \pi') \neq (U, \pi).
\end{aligned} \tag{B.54}$$

One can easily show that $P_{(k)}$ ($k = 1, 2$) satisfy the detailed balance condition with respect to molecular dynamics, as in the flat case:

$$P_{(k)}(U', \pi' | U, \pi) e^{-H(U, \pi)} = P_{(k)}(U, -\pi | U', -\pi') e^{-H(U', -\pi')}. \tag{B.55}$$

Although neither $P_{(1)}$ nor $P_{(2)}$ is ergodic,³⁴ their product $P \equiv P_{(2)}P_{(1)}$ is expected to be ergodic to guarantee unique convergence to the equilibrium distribution, and thus the Markov chain defined by P will give $e^{-H(U, \pi)} / \int_{TG} d\Omega e^{-H(U, \pi)}$ as the equilibrium distribution.³⁵

³⁴ $P_{(1)}$ keeps U fixed, while $P_{(2)}$ evolves the system along (approximately) constant energy surfaces.

³⁵Although not necessary, one can verify that the induced kernel

$$P(U' | U) \equiv \int \left(\prod_a d\pi'_a \right) \left(\prod_a d\pi_a \right) P(U', \pi' | U, \pi) \tag{B.56}$$

satisfies detailed balance with respect to $e^{-V(U)}$,

$$P(U' | U) e^{-V(U)} = P(U | U') e^{-V(U')}. \tag{B.57}$$

Since the observables depend only on U , the algorithm can be expressed as a stochastic process for U as follows [18]:

- Step 1 (momentum refresh):
Given $U \in G$, generate $\pi \in T_U G$ from the Gaussian distribution $\propto e^{-(1/2) \text{tr} \pi^\dagger \pi}$.
- Step 2 (molecular dynamics):
Evolve $(U, \pi) \rightarrow (U', \pi')$ using the MD algorithm [Eq. (B.35)].
- Step 3 (Metropolis test):
Accept the proposed U' with probability $\min(1, e^{-[H(U', \pi') - H(U, \pi)]})$.

C. Random generation of elements of $\text{Lie } G$ and $\text{Lie } G^\mathbb{C}$

Let $\{T_a\}$ ($a = 1, \dots, N (= \dim G)$) be a basis of the Lie algebra $\text{Lie } G$, satisfying $T_a^\dagger = -T_a$ and $\text{tr}(T_a T_b) = -\delta_{ab}$. We present algorithms for generating random elements $X = T_a x^a \in \text{Lie } G$ ($x^a \in \mathbb{R}$) according to the Gaussian distribution $e^{-\langle X, X \rangle / (2\sigma^2)} = e^{-(x^a)^2 / (2\sigma^2)}$,

$$\langle x^a \rangle = 0, \quad \langle x^a x^b \rangle = \sigma^2 \delta^{ab}, \quad (\text{C.1})$$

and also for generating random element $Z = T_a z^a \in \text{Lie } G^\mathbb{C}$ ($z^a \in \mathbb{C}$) according to the Gaussian distribution $e^{-\langle Z, Z \rangle / (2\sigma^2)} = e^{-\bar{z}^a z^a / (2\sigma^2)}$,

$$\langle z^a \rangle = 0, \quad \langle z^a z^b \rangle = \langle \bar{z}^a \bar{z}^b \rangle = 0, \quad \langle z^a \bar{z}^b \rangle = 2\sigma^2 \delta^{ab}. \quad (\text{C.2})$$

A direct algorithm is to generate x^a and y^a ($a = 1, \dots, N$) from $\propto e^{-(x^a)^2 / (2\sigma^2)}$ and $\propto e^{-(y^a)^2 / (2\sigma^2)}$, respectively, and set $X = T_a x^a$ and $Z = T_a(x^a + iy^a)$. In the following, we present an algorithm that does not use the explicit form of T_a . Throughout this section, we restrict our discussions to the cases where $G = SU(n)$ or $G = U(n)$. Other classical compact groups can be treated similarly.

C.1. $G = SU(n)$

In this case, $N = \dim G = n^2 - 1$. A random variable $X = T_a x^a \in \mathfrak{su}(n) = \text{Lie } SU(n)$ satisfying Eq. (C.1) can be constructed by the following two steps:

(1): Generate two sets of independent Gaussian random variables $\xi = (\xi_{ij})$ and $\eta = (\eta_{ij})$,

$$\xi_{ij} \sim e^{-\xi_{ij}^2 / (2\sigma^2)}, \quad \eta_{ij} \sim e^{-\eta_{ij}^2 / (2\sigma^2)} \quad (i, j = 1, \dots, N). \quad (\text{C.3})$$

(2): Set $X = (X_{ij})$ as

$$X = \frac{1}{2} (\xi - \xi^T) + \frac{i}{2} (\eta + \eta^T) - \frac{i}{n} (\text{tr } \eta) 1, \quad (\text{C.4})$$

where 1 is the $n \times n$ identity matrix.

To verify that this construction yields the desired distribution, note that x^a are given by $x^a = -\text{tr}(T_a X) = \text{tr}(T^a X) = T_{ji}^a X_{ij}$. Thus, the first moment becomes

$$\langle x^a \rangle = T_{ji}^a \langle X_{ij} \rangle = T_{ji}^a \times \langle (\text{terms linear in } \xi \text{ and } \eta) \rangle = 0. \quad (\text{C.5})$$

For the second moment, note that $\langle X_{ij} X_{kl} \rangle$ in

$$\langle x^a x^b \rangle = T_{ji}^a T_{lk}^b \langle X_{ij} X_{kl} \rangle \quad (\text{C.6})$$

can be computed as

$$\begin{aligned} \langle X_{ij} X_{kl} \rangle &= \frac{1}{4} \left[\langle (\xi_{ij} - \xi_{ji})(\xi_{kl} - \xi_{lk}) \rangle - \left\langle \left(\eta_{ij} + \eta_{ji} - \frac{2}{n} \eta_{pp} \delta_{ij} \right) \left(\eta_{kl} + \eta_{lk} - \frac{2}{n} \eta_{qq} \delta_{kl} \right) \right\rangle \right] \\ &= -\sigma^2 \left(\delta_{il} \delta_{jk} - \frac{1}{n} \delta_{ij} \delta_{kl} \right), \end{aligned} \quad (\text{C.7})$$

so that we have

$$\langle x^a x^b \rangle = -\sigma^2 \text{tr}(T^a T^b) = \sigma^2 \delta^{ab}. \quad (\text{C.8})$$

C.2. $G^{\mathbb{C}} = SU(n)^{\mathbb{C}} = SL(n, \mathbb{C})$

In this case, $N = \dim G = n^2 - 1$. A random variable $Z = T_a z^a \in \mathfrak{sl}(n, \mathbb{C}) = \text{Lie } SL(n, \mathbb{C})$ satisfying Eq. (C.2) can be generated by the following two steps:

(1): Generate two sets of independent Gaussian random variables $\xi = (\xi_{ij})$ and $\eta = (\eta_{ij})$,

$$\xi_{ij} \sim e^{-\xi_{ij}^2/(2\sigma^2)}, \quad \eta_{ij} \sim e^{-\eta_{ij}^2/(2\sigma^2)} \quad (i, j = 1, \dots, N). \quad (\text{C.9})$$

(2): Set $Z = (Z_{ij})$ as

$$Z = \xi + i\eta - \frac{1}{n} (\text{tr } \xi + i \text{tr } \eta) 1, \quad (\text{C.10})$$

where 1 is the $n \times n$ identity matrix.

To verify that this construction yields the desired distribution, note that z^a and $\bar{z}^b$ are expressed as $z^a = -\text{tr}(T_a Z) = \text{tr}(T^a Z) = T_{ji}^a Z_{ij}$ and $\bar{z}^b = \text{tr}(T^{b\dagger} Z^\dagger) = -\text{tr}(T^b Z^\dagger) = -T_{kl}^b \overline{Z_{kl}}$. Thus, the first moment becomes

$$\langle z^a \rangle = T_{ji}^a \langle Z_{ij} \rangle = T_{ji}^a \times \langle (\text{terms linear in } \xi \text{ and } \eta) \rangle = 0. \quad (\text{C.11})$$

For the second moment, note that $\langle Z_{ij} Z_{kl} \rangle$ in

$$\langle z^a z^b \rangle = T_{ji}^a T_{lk}^b \langle Z_{ij} Z_{kl} \rangle \quad (\text{C.12})$$

can be computed as

$$\begin{aligned}\langle Z_{ij} Z_{kl} \rangle &= \left\langle \left(\xi_{ij} - \frac{1}{n} \xi_{pp} \delta_{ij} \right) \left(\xi_{kl} - \frac{1}{n} \xi_{qq} \delta_{kl} \right) \right\rangle \\ &\quad - \left\langle \left(\eta_{ij} - \frac{1}{n} \eta_{pp} \delta_{ij} \right) \left(\eta_{kl} - \frac{1}{n} \eta_{qq} \delta_{kl} \right) \right\rangle \\ &= 0.\end{aligned}\tag{C.13}$$

Furthermore, $\langle Z_{ij} \overline{Z_{kl}} \rangle$ in

$$\langle z^a \overline{z^b} \rangle = -T_{ji}^a T_{kl}^b \langle Z_{ij} \overline{Z_{kl}} \rangle\tag{C.14}$$

can be computed as

$$\langle Z_{ij} \overline{Z_{kl}} \rangle = 2\sigma^2 \left(\delta_{ik} \delta_{jl} - \frac{1}{n} \delta_{ij} \delta_{kl} \right),\tag{C.15}$$

so that we have

$$\langle z^a \overline{z^b} \rangle = -2\sigma^2 \text{tr} (T^a T^b) = 2\sigma^2 \delta^{ab}.\tag{C.16}$$

C.3. $G = U(n)$

In this case, $N = \dim G = n^2$. A random variable $X = T_a x^a \in \mathfrak{u}(n) = \text{Lie } U(n)$ satisfying Eq. (C.1) can be constructed by the following two steps:

(1): Generate two sets of independent Gaussian random variables $\xi = (\xi_{ij})$ and $\eta = (\eta_{ij})$,

$$\xi_{ij} \sim e^{-\xi_{ij}^2/(2\sigma^2)}, \quad \eta_{ij} \sim e^{-\eta_{ij}^2/(2\sigma^2)} \quad (i, j = 1, \dots, N).\tag{C.17}$$

(2): Set $X = (X_{ij})$ as

$$X = \frac{1}{2} (\xi - \xi^T) + \frac{i}{2} (\eta + \eta^T).\tag{C.18}$$

C.4. $G^{\mathbb{C}} = U(n)^{\mathbb{C}} = GL(n, \mathbb{C})$

In this case, $N = \dim G = n^2$. A random variable $Z = T_a z^a \in \mathfrak{gl}(n, \mathbb{C}) = \text{Lie } GL(n, \mathbb{C})$ satisfying Eq. (C.2) can be constructed by the following two steps:

(1): Generate two sets of independent Gaussian random variables $\xi = (\xi_{ij})$ and $\eta = (\eta_{ij})$,

$$\xi_{ij} \sim e^{-\xi_{ij}^2/(2\sigma^2)}, \quad \eta_{ij} \sim e^{-\eta_{ij}^2/(2\sigma^2)} \quad (i, j = 1, \dots, N).\tag{C.19}$$

(2): Set $Z = (Z_{ij})$ as

$$Z = \xi + i\eta.\tag{C.20}$$

D. Proof of Eqs. (4.39)–(4.42)

Equations (4.39), (4.40) and (4.42) follow directly from the Poisson brackets (4.16)–(4.24), as shown below.

$$\bullet \quad T_{V/2} U = e^{(\epsilon/2) \{*, V\}} U = U, \quad (\text{D.1})$$

$$\bullet \quad T_{V/2} \pi = e^{(\epsilon/2) \{*, V\}} \pi = \pi + \frac{\epsilon}{2} \{ \pi, V \} = \pi - \epsilon (DV)^\dagger, \quad (\text{D.2})$$

$$\bullet \quad T_K \pi = e^{\epsilon \{*, K\}} \pi = e^{\epsilon [\pi - \pi^\dagger, *]} \pi = e^{\epsilon (\pi - \pi^\dagger)} \pi e^{-\epsilon (\pi - \pi^\dagger)}. \quad (\text{D.3})$$

In order to show Eq. (4.41), we notice that $T_K U$ can be rewritten to the form

$$T_K U = f(\pi, \pi^\dagger) U \equiv \left[\sum_{n=0}^{\infty} \frac{\epsilon^n}{n!} f_n(\pi, \pi^\dagger) \right] U. \quad (\text{D.4})$$

Here, the coefficients $f_n(\pi, \pi^\dagger)$ are determined recursively with $f_0 = 1$ as follows:

$$f_{n+1} = [\pi - \pi^\dagger, f_n] + f_n \pi, \quad (\text{D.5})$$

because

$$f_{n+1} U = \{f_n U, K\} = \{f_n, K\} U + f_n \{U, K\} = ([\pi - \pi^\dagger, f_n] + f_n \pi) U, \quad (\text{D.6})$$

where we have used the formula (4.37). The right-hand side of Eq. (D.5) can be further rewritten as

$$(\pi - \pi^\dagger) f_n + f_n \pi^\dagger = (L_{\pi - \pi^\dagger} + R_{\pi^\dagger}) f_n, \quad (\text{D.7})$$

where L_X and R_X denote multiplications with matrix X from left and right, respectively,

$$L_X f \equiv X f, \quad R_X f \equiv f X, \quad (\text{D.8})$$

which commute with each other, $L_X R_Y = R_Y L_X$ ($\forall X, Y$). We thus have

$$f_n(\pi, \pi^\dagger) = (L_{\pi - \pi^\dagger} + R_{\pi^\dagger})^n 1, \quad (\text{D.9})$$

which leads to

$$\sum_{n=0}^{\infty} \frac{\epsilon^n}{n!} f_n(\pi, \pi^\dagger) = e^{L_{\pi - \pi^\dagger} + R_{\pi^\dagger}} 1 = e^{L_{\pi - \pi^\dagger}} e^{R_{\pi^\dagger}} 1 = e^{\pi - \pi^\dagger} e^{\pi^\dagger}. \quad (\text{D.10})$$

We thus have derived the desired equation [Eq. (4.41)]:

$$\bullet \quad T_K U = e^{\pi - \pi^\dagger} e^{\pi^\dagger} U \quad (\text{D.11})$$

E. Proof of Eq. (7.16)

We start from the expression

$$(DV)^\dagger = \frac{1}{2} (DS)^\dagger + W'(t) (Dt)^\dagger = \frac{1}{2} \xi + W'(t) (Dt)^\dagger. \quad (\text{E.1})$$

We decompose $(Dt)^\dagger \in T_U G^\mathbb{C}$ at $U \in \Sigma_t (\subset \mathcal{R})$ into the form

$$(Dt)^\dagger = (Dt)^\dagger_{\parallel} + (Dt)^\dagger_{\perp} = v + c \xi_n + (Dt)^\dagger_{\perp} \quad (\text{E.2})$$

with $v = E_a v^a \in T_U \Sigma_t (\subset T_U \mathcal{R})$. Note that $\xi_n \in N_U \Sigma_t \cap T_U \mathcal{R}$ and $(Dt)^\dagger_{\perp} \in N_U \mathcal{R}$. In the following, we will show that (1) $v = 0$ and (2) $c = 1/(2 \langle \xi_n, \xi_n \rangle)$. This completes the proof because $(Dt)^\dagger_{\perp} \in N_U \mathcal{R}$ can be absorbed into λ in Eq. (7.12) and $(Dt')^\dagger_{\perp} \in N_{U'} \mathcal{R}$ drops under the projection Eq. (7.14).

(1) For $\forall U \in \Sigma_t, \forall u = E_a u^a \in T_U \Sigma_t$ and an infinitesimally small ϵ , we have

$$t(U, U^\dagger) = t(e^{\epsilon u} U, (e^{\epsilon u} U)^\dagger) = t(U, U^\dagger) + \epsilon \text{tr} [u Dt + u^\dagger (Dt)^\dagger] + O(\epsilon^2), \quad (\text{E.3})$$

and thus,

$$0 = \text{tr} [u Dt + u^\dagger (Dt)^\dagger] = 2 \langle u, (Dt)^\dagger \rangle = 2 \langle u, v \rangle = 2 u^a \gamma_{ab} v^b. \quad (\text{E.4})$$

This means that $v = 0$ due to the nondegeneracy of γ_{ab} .

(2) By using the orthogonality, c is given by

$$c = \frac{\langle \xi_n, (Dt)^\dagger \rangle}{\langle \xi_n, \xi_n \rangle}. \quad (\text{E.5})$$

Here, noting that

$$\begin{aligned} 1 &= [t(U, U^\dagger)]^\bullet = \text{tr} [(\dot{U} U^{-1}) Dt + (\dot{U} U^{-1})^\dagger (Dt)^\dagger] \\ &= 2 \langle \xi, (Dt)^\dagger \rangle = 2 \langle \xi_n, (Dt)^\dagger \rangle \quad (\because v = 0), \end{aligned} \quad (\text{E.6})$$

we have $\langle \xi_n, (Dt)^\dagger \rangle = 1/2$. We thus obtain $c = 1/(2 \langle \xi_n, \xi_n \rangle)$.

References

- [1] E. Witten, “Analytic continuation of Chern-Simons theory,” AMS/IP Stud. Adv. Math. **50**, 347-446 (2011) [arXiv:1001.2933 [hep-th]].
- [2] M. Cristoforetti, F. Di Renzo and L. Scorzato, “New approach to the sign problem in quantum field theories: High density QCD on a Lefschetz thimble,” Phys. Rev. D **86**, 074506 (2012) [arXiv:1205.3996 [hep-lat]].

- [3] M. Cristoforetti, F. Di Renzo, A. Mukherjee and L. Scorzato, “Monte Carlo simulations on the Lefschetz thimble: Taming the sign problem,” *Phys. Rev. D* **88**, no. 5, 051501(R) (2013) [arXiv:1303.7204 [hep-lat]].
- [4] H. Fujii, D. Honda, M. Kato, Y. Kikukawa, S. Komatsu and T. Sano, “Hybrid Monte Carlo on Lefschetz thimbles - A study of the residual sign problem,” *JHEP* **1310**, 147 (2013) [arXiv:1309.4371 [hep-lat]].
- [5] H. Fujii, S. Kamata and Y. Kikukawa, “Lefschetz thimble structure in one-dimensional lattice Thirring model at finite density,” *JHEP* **11**, 078 (2015) [erratum: *JHEP* **02**, 036 (2016)] [arXiv:1509.08176 [hep-lat]].
- [6] H. Fujii, S. Kamata and Y. Kikukawa, “Monte Carlo study of Lefschetz thimble structure in one-dimensional Thirring model at finite density,” *JHEP* **12**, 125 (2015) [erratum: *JHEP* **09**, 172 (2016)] [arXiv:1509.09141 [hep-lat]].
- [7] A. Alexandru, G. Başar and P. Bedaque, “Monte Carlo algorithm for simulating fermions on Lefschetz thimbles,” *Phys. Rev. D* **93**, no. 1, 014504 (2016) [arXiv:1510.03258 [hep-lat]].
- [8] A. Alexandru, G. Başar, P. F. Bedaque, G. W. Ridgway and N. C. Warrington, “Sign problem and Monte Carlo calculations beyond Lefschetz thimbles,” *JHEP* **1605**, 053 (2016) [arXiv:1512.08764 [hep-lat]].
- [9] A. Alexandru, G. Başar, P. F. Bedaque and G. W. Ridgway, “Schwinger-Keldysh formalism on the lattice: A faster algorithm and its application to field theory,” *Phys. Rev. D* **95**, no.11, 114501 (2017) [arXiv:1704.06404 [hep-lat]].
- [10] M. Fukuma and N. Umeda, “Parallel tempering algorithm for integration over Lefschetz thimbles,” *PTEP* **2017**, no. 7, 073B01 (2017) [arXiv:1703.00861 [hep-lat]].
- [11] A. Alexandru, G. Başar, P. F. Bedaque and N. C. Warrington, “Tempered transitions between thimbles,” *Phys. Rev. D* **96**, no. 3, 034513 (2017) [arXiv:1703.02414 [hep-lat]].
- [12] M. Fukuma, N. Matsumoto and N. Umeda, “Applying the tempered Lefschetz thimble method to the Hubbard model away from half filling,” *Phys. Rev. D* **100**, no. 11, 114510 (2019) [arXiv:1906.04243 [cond-mat.str-el]].
- [13] A. Alexandru, “Improved algorithms for generalized thimble method,” talk at the 37th international conference on lattice field theory, Wuhan, 2019.
- [14] M. Fukuma, N. Matsumoto and N. Umeda, “Implementation of the HMC algorithm on the tempered Lefschetz thimble method,” [arXiv:1912.13303 [hep-lat]].
- [15] M. Fukuma and N. Matsumoto, “Worldvolume approach to the tempered Lefschetz thimble method,” *PTEP* **2021**, no.2, 023B08 (2021) [arXiv:2012.08468 [hep-lat]].

- [16] M. Fukuma, N. Matsumoto and Y. Namekawa, “Statistical analysis method for the worldvolume hybrid Monte Carlo algorithm,” PTEP **2021**, no.12, 123B02 (2021) [arXiv:2107.06858 [hep-lat]].
- [17] M. Fukuma, “Simplified Algorithm for the Worldvolume HMC and the Generalized Thimble HMC,” PTEP **2024**, no.5, 053B02 (2024) [arXiv:2311.10663 [hep-lat]].
- [18] S. Duane, A. D. Kennedy, B. J. Pendleton and D. Roweth, “Hybrid Monte Carlo,” Phys. Lett. B **195**, 216-222 (1987)
- [19] M. Fukuma and Y. Namekawa, “Applying the Worldvolume Hybrid Monte Carlo method to the complex scalar field theory at finite density,” in preparation.
- [20] H. C. Andersen, “RATTLE: A “velocity” version of the SHAKE algorithm for molecular dynamics calculations,” J. Comput. Phys. **52**, 24 (1983).
- [21] B. J. Leimkuhler and R. D. Skeel, “Symplectic numerical integrators in constrained Hamiltonian systems,” J. Comput. Phys. **112**, 117 (1994).
- [22] H. Munthe-Kaas, “Lie-Butcher theory for Runge-Kutta methods,” BIT **35** 572 (1995).
- [23] H. Munthe-Kaas, “Runge-Kutta methods on Lie groups,” BIT **38** 92 (1998).